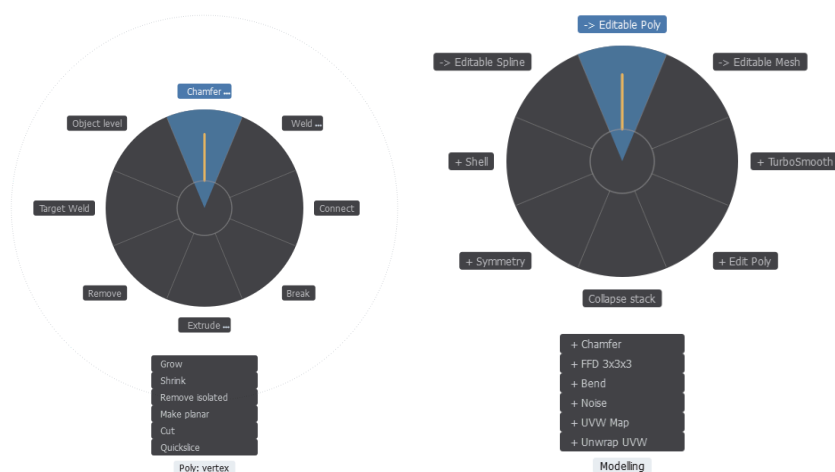


MarkingForge

QuickStart - from your first gesture to your own dials

3DS MAX 2027 · VERSION 1.5.0 · ABOUT 40 MINUTES



REAL MARKING MENUS - A QUICK GESTURE NEEDS NO DIAL

MarkingForge works the way marking menus are meant to: the command is chosen by the **direction** of the movement, not by clicking an item. Press the key and flick the mouse towards the command in one quick movement - the command runs and **the dial is not even drawn**. The dial appears only when you hold the key and wait, to remind you where things are. After a few days your hand knows the directions and you stop looking.

Twenty-two short tutorials. Part 1 teaches your hand the dials that come with MarkingForge. Part 2 teaches the menu editor: putting your own commands on a dial, context variants, submenus, scripts, colours and ready-made looks, shortcuts, pages of one dial under the mouse wheel, and the dial library. Every picture is the real program. The **Installation Guide** covers installing; the **Reference** explains every option in detail.

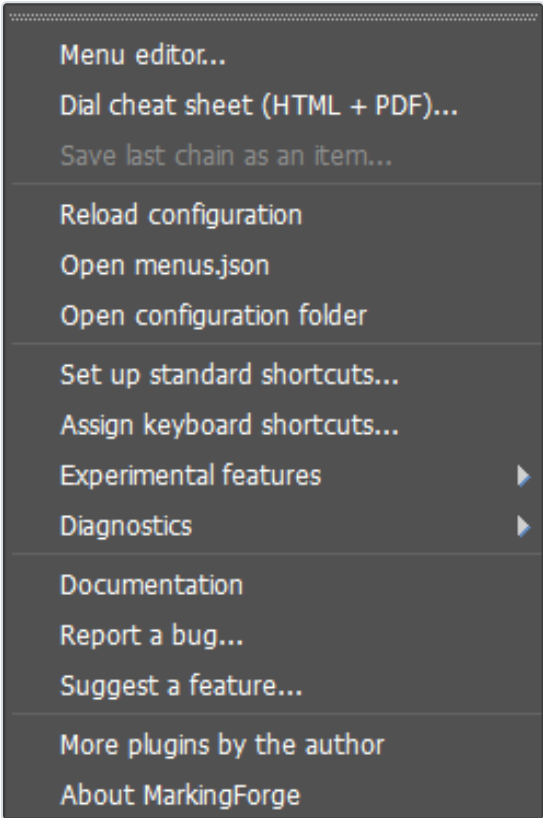
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Before you begin

Check two things - it takes ten seconds:

- 1 There is a **MarkingForge** menu in the 3ds Max menu bar. If not, see the Installation Guide.
- 2 The dials have keys. On its first start MarkingForge gives every dial its standard key by itself - Alt, Ctrl+Alt and Shift+Alt with 1-8 - when no dial has a key yet, and skips a key something else already uses. To check, open **MarkingForge > Set up standard shortcuts...**: it either lists the keys it will give, or says every free slot is already set up.



The MarkingForge menu. Everything in this guide starts here or on a key.

Your 40 minutes

Minutes	What you do
0 - 10	Tutorials 1-6: the gestures and the built-in dials.
10 - 15	Tutorial 7: the editor's layout. Tutorial 8: your first own command on a dial.
15 - 40	Tutorials 9-22: labels, lists, shortcuts, context variants, submenus, scripts, colours and ready-made looks, the editor's own colours, sharing, dials that fill themselves, pages under the mouse wheel, the dial library, the new-dial wizard.

Six ideas in one minute

1. A dial is a round menu under a key

Hold Alt+1 and eight commands appear around the cursor. 24 dials sit on Alt, Ctrl+Alt and Shift+Alt with the digits 1-8.

2. You aim, you do not read

Move towards a command and let go of the key. The direction is what your hand remembers - after a week you will not look.

3. Hold, flick or tap

Hold to see the dial. Flick - press and move at once - to run a command without the dial ever appearing. Release in the centre to cancel.

4. The dial follows your work

The same key shows vertex tools at vertex level, edge tools at edge level, spline tools on a shape - and material tools over the Slate editor.

5. Everything can be changed

The menu editor puts any of 3ds Max's 4000+ commands, a script, a submenu or a slider on any direction. One key can even hold several dials - pages you turn with the mouse wheel. Nothing needs a restart.

6. It is safe to try

Cancel is always one move away, every command undoes with Ctrl+Z, and the editor changes nothing until you press Save.

PART 1

Using the dials

Six tutorials that teach your hand the dials MarkingForge comes with. Have an empty scene open.

1

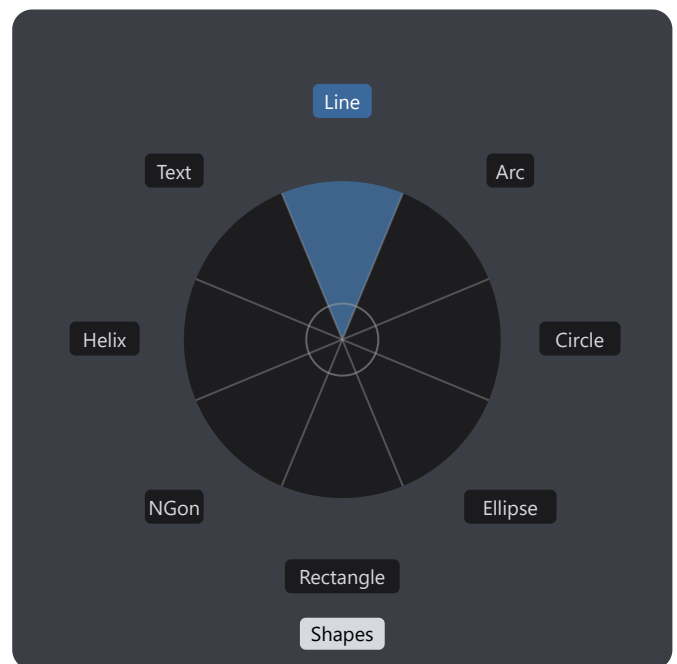
Your first gesture

2 min · You will learn: to hold, aim and release

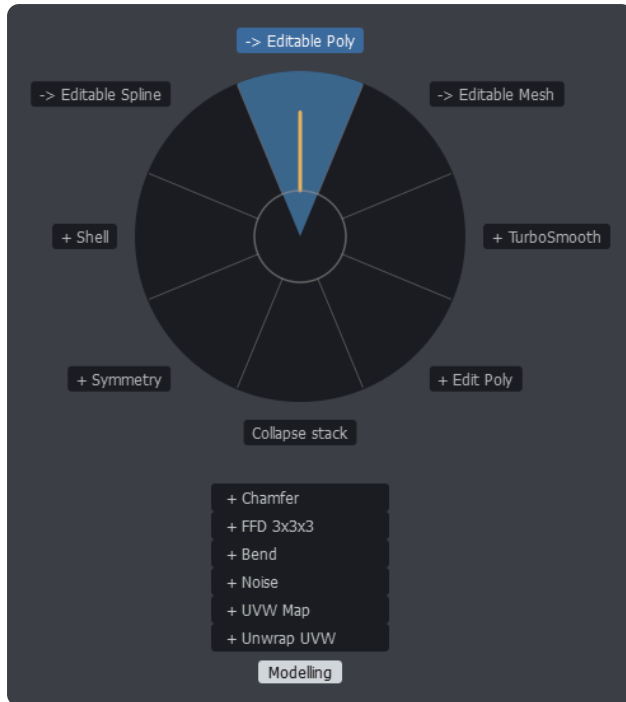
- 1 Move the mouse over a viewport.
- 2 Hold **Alt+3**. The **Create** dial appears around the cursor.
- 3 Move the mouse up - towards **Box**. The wedge lights up.
- 4 Release the keys. 3ds Max is now in Box creation: drag a box in the viewport as usual.
- 5 Hold **Alt+3** again and move **down**, through **Shapes >**, without releasing: a second dial opens where the cursor is - a **submenu**. Move up to **Line** and release. Esc ends the line.
- 6 Click the box to select it, hold **Alt+1** (Modelling) over it, move up to -> **Editable Poly** and release.



Alt+3 - Create: four solids from north to south-east; four groups from south to north-west, each a submenu (marked >).



...through Shapes: the Shapes submenu, aimed at Line.



Alt+1 aimed north: the box becomes an Editable Poly.

RESULT

The box is an Editable Poly - look at the modifier stack. Two gestures, no menu, no command panel.

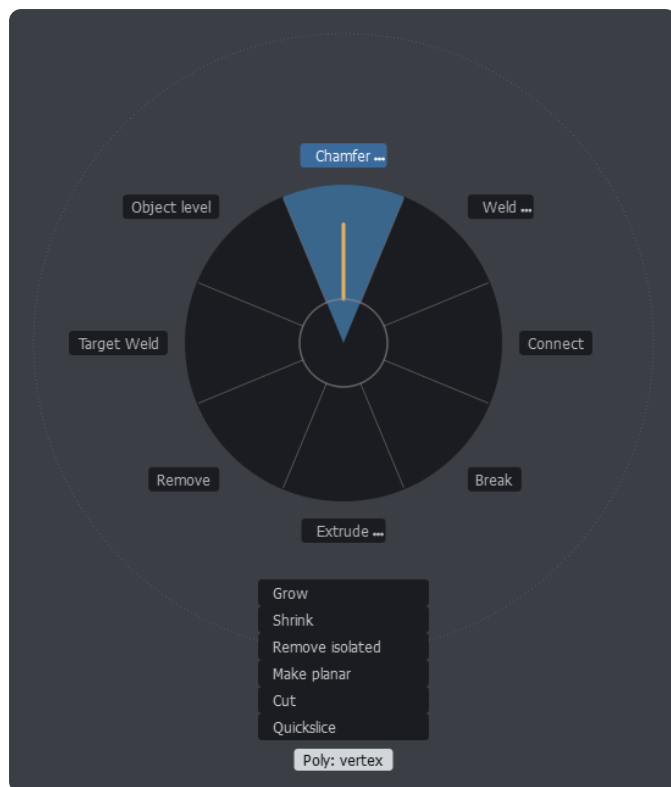
Changed your mind? Move back to the centre and release, or press Esc. Nothing runs.

2

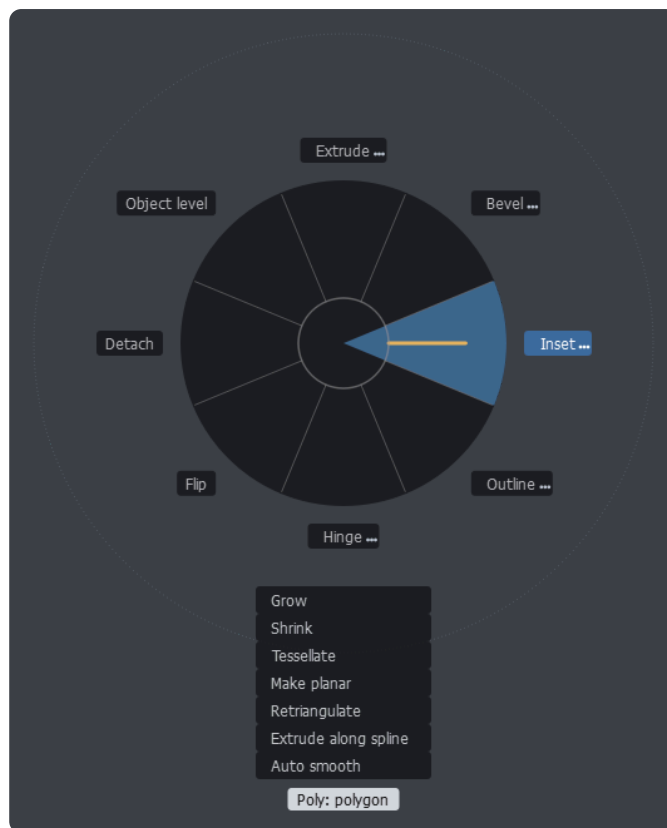
The dial follows you

4 min · You will learn: how one key changes with the sub-object level

- 1 Keep the Editable Poly selected and press **1** (vertex level).
- 2 Hold **Alt+1**: the same key now shows vertex tools - Chamfer, Weld, Connect, Break...
- 3 Aim at **Chamfer** (north) and release, then drag in the viewport.
- 4 Press **2** (edge level), select an edge and hold **Alt+1**: Connect, Chamfer, Loop, Ring...
- 5 Press **4** (polygon level), select a polygon, hold **Alt+1**, aim east - **Inset**.
- 6 Aim **north-west** at any level - **Object level** takes you back up.



Vertex level, aimed at Chamfer.



Polygon level, aimed at Inset.

One layout for both kinds of poly. Chamfer, Extrude, Weld... sit in the same direction on an Editable Poly and under an Edit Poly modifier. Learn it once.

RESULT

You modelled at three sub-object levels with one key.

3

Flick - the fast way

3 min - You will learn: to run commands without waiting for the dial

A marking menu is fastest when you do not wait for it. Press the key and move in the same instant, then release: the command runs, the dial is never drawn.

TRY IT

Repeat each gesture five times, slowly at first, then without looking at the dial:

Gesture	Does
Alt+4, up	Zoom extents all
Alt+4, down	Isolate the selection (again = end isolate)
Alt+5, up / down	Select all / select none
Ctrl+Alt+1, up / right	Move / scale tool
Alt+7, up	Undo one step

RESULT

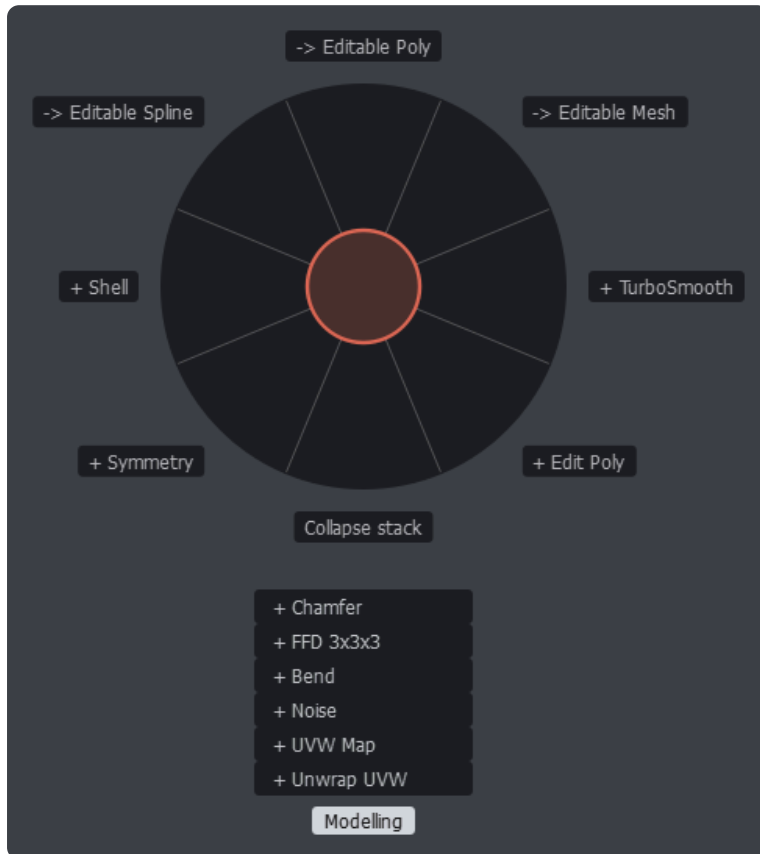
After a day these five need no looking. That is the whole idea - the rest of the dials follow the same way.

4

More than eight - the list and the settings rim

3 min · You will learn: the rows under a dial and opening a command's settings

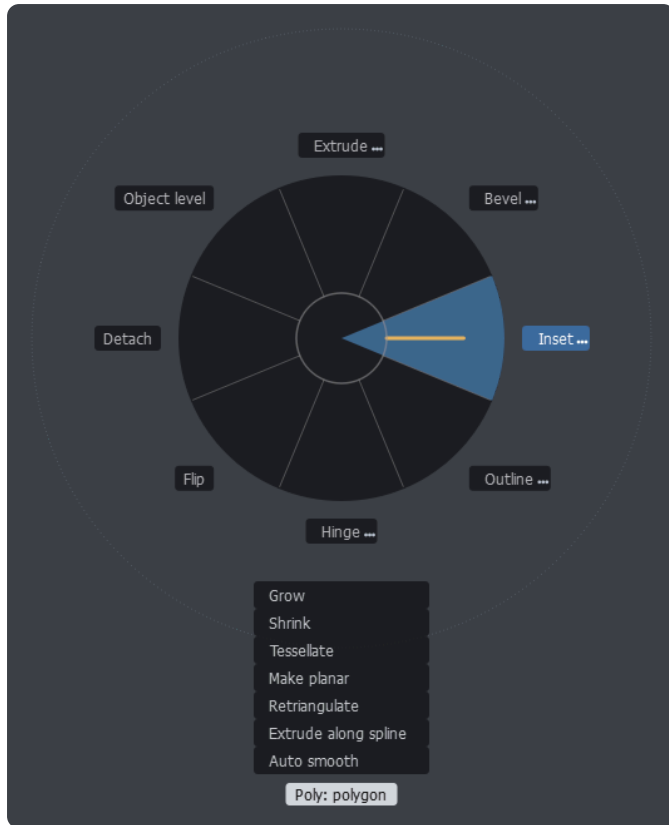
Some dials show **rows under the ring**: commands you use, but not often enough to spend a direction on. Keep holding the key, move down to a row and click it.



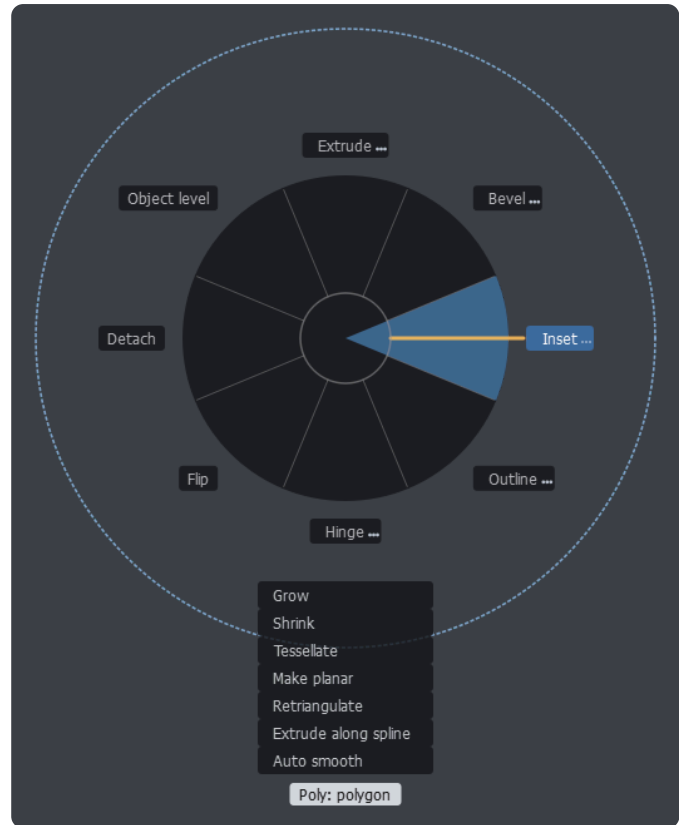
Alt+1 on an ordinary object: eight directions plus six rows (Chamfer, FFD, Bend, Noise, UVW Map, Unwrap UVW).

Commands with a settings window (Chamfer, Extrude, Inset...) carry ... on their tile. An ordinary release **opens that window**, like the settings icon in a quad menu. Move **far past the ring** before releasing and the command runs **without** it - with the last values.

Where "far" begins is drawn on the dial: the **dotted circle outside the ring** is the settings rim. Cross it and the circle **brightens**, and the dots on the tile dim - that is the dial telling you the window will be skipped. Come back inside the circle before releasing and the window opens after all.



Inside the dotted circle: Inset will open its settings.



Past it: the circle lights up - Inset runs without the window.

TRY IT

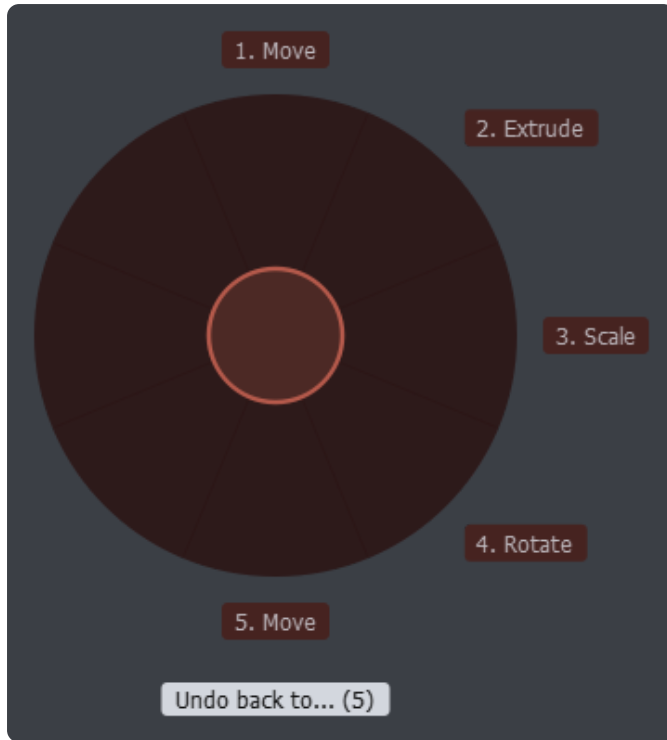
On a polygon, hold **Alt+1**, aim east (Inset) and release: the Inset settings open. Do it again, but move far out before releasing: Inset runs straight away with the same amount.

Prefer the opposite for a command you repeat a lot? Select its direction in the editor and press **Settings...** (tutorial 7 shows where).

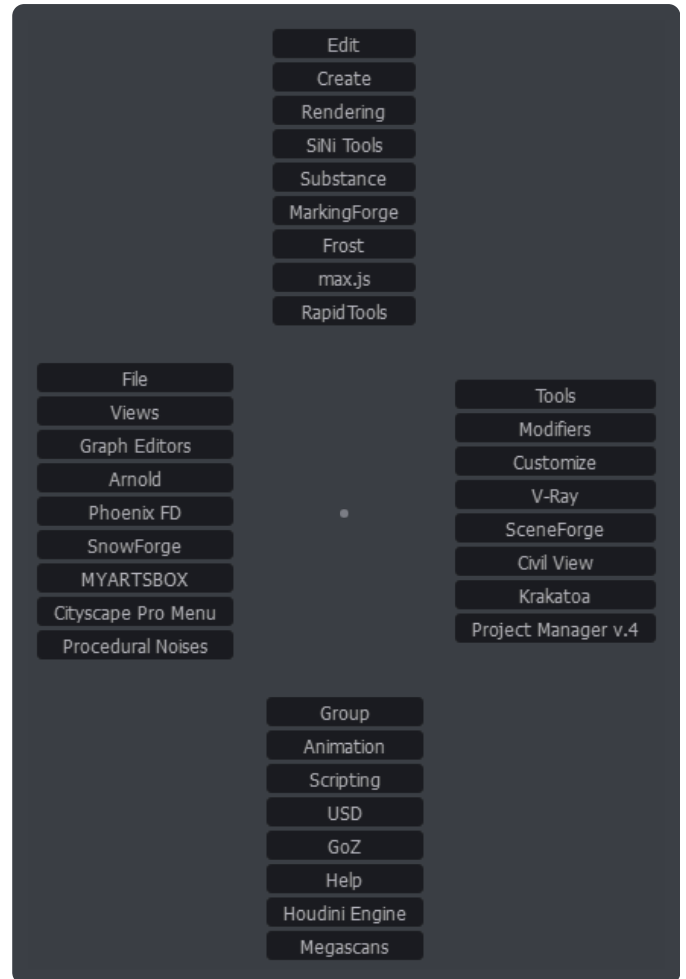
5 Undo by name, repeat, and the hotbox

3 min · You will learn: three dials that fill themselves

- 1 **Alt+7** - the undo dial shows your last eight operations by name. North undoes one, north-east two, and so on: undo three steps in one move.
- 2 **Alt+8** - recent commands: what you ran from the dials, newest first. Repeat the last one with a flick up.
- 3 **Alt+2**, then one notch of the mouse wheel **away** from you - the hotbox: every menu of 3ds Max as tiles, including menus other plugins add. Point at a title, then at an entry, and release - or click the entry with the left mouse button. (**Alt+2** itself opens the viewport dials - shading, lighting, display and views, one page each.)



Alt+7 - undo by name.



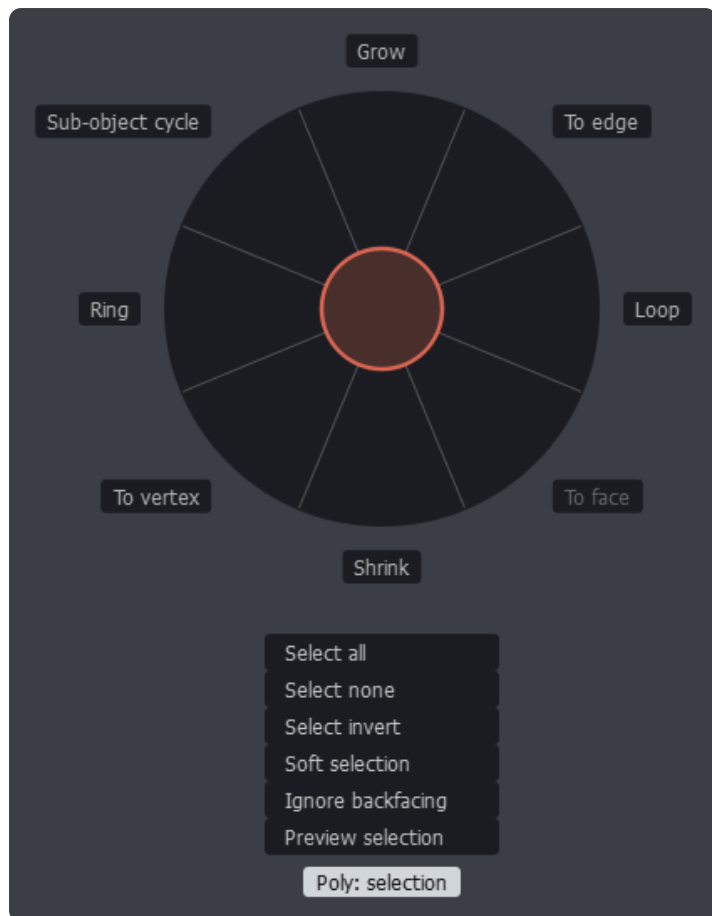
Alt+2, one wheel notch away - the hotbox.

The **modifier stack** dial (Alt+6) works the same way: the top eight modifiers of the selected object - aim at one to make it the current one in the Modify panel.

6 Selection and hiding on sub-objects

2 min · You will learn: context on other dials

- 1 At polygon level hold **Alt+5** (Selection): Grow and Shrink face each other, Loop and Ring face each other; on an Editable Poly three corners convert the selection to vertices, edges or faces (under an Edit Poly modifier they are Select all, Select invert and Select none).
- 2 Hold **Ctrl+Alt+3** (Show and hide) at the same level: hide or unhide polygons, not whole objects.
- 3 Return to object level (Alt+1, north-west) and hold the same keys again - they work on whole objects.



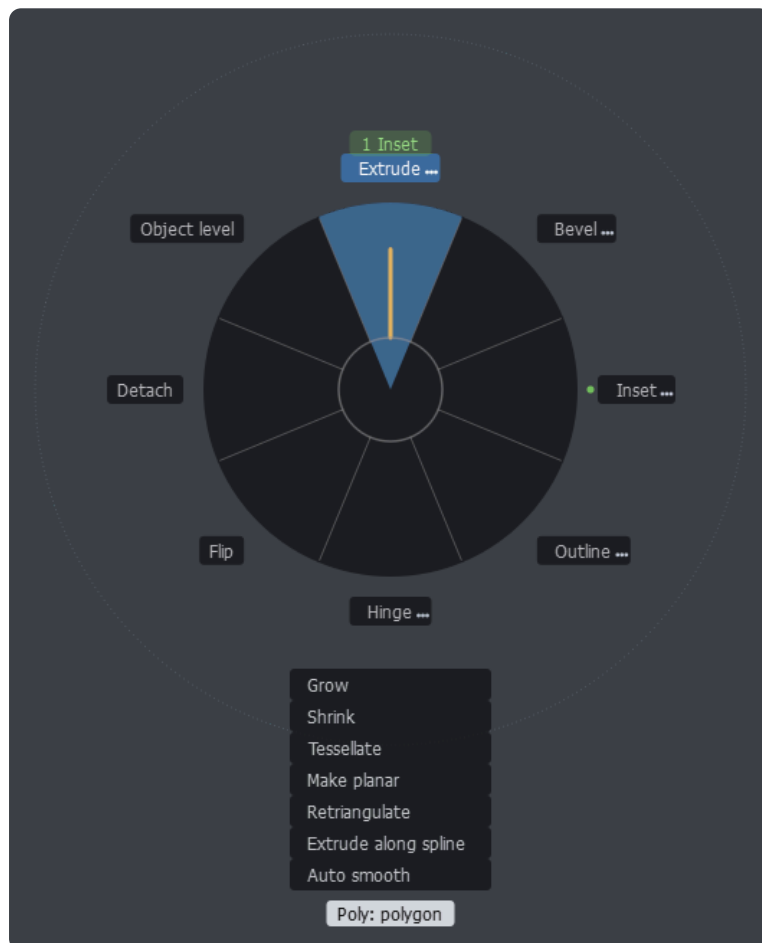
Alt+5 at a sub-object level.

Five tricks worth knowing

Tricks 2-4 work from the first start; the first and the fifth have to be switched on. Each feature behind them has its tick in **MarkingForge > Experimental features**, so any of them can be turned off.

1. Several commands in one gesture (switch it on first)

- 1 Switch on **Multiple picks in one gesture** in **MarkingForge > Experimental features**.
- 2 Hold the key and move out **past the ring** towards a command.
- 3 **Come back to the dashed circle** round the centre without releasing: the command is queued, and the queue appears as a numbered row above the dial (*1 Inset > 2 Extrude*).
- 4 Queue more the same way, then release inside the dashed circle (it is green): the whole queue runs in order, as **one undo step**.
- 5 Changed your mind? Release at the very centre (the small dot) or over an empty direction - the queue is discarded.



Inset is queued (the row above the dial); the hand is out at Extrude. The dashed circle round the centre is where "back" is.

Example: at polygon level, Alt+1 - out to **Inset** and back, then out to **Extrude** and release. One Ctrl+Z takes both back. A chain you repeat can become one direction: **MarkingForge > Save last chain as an item...**

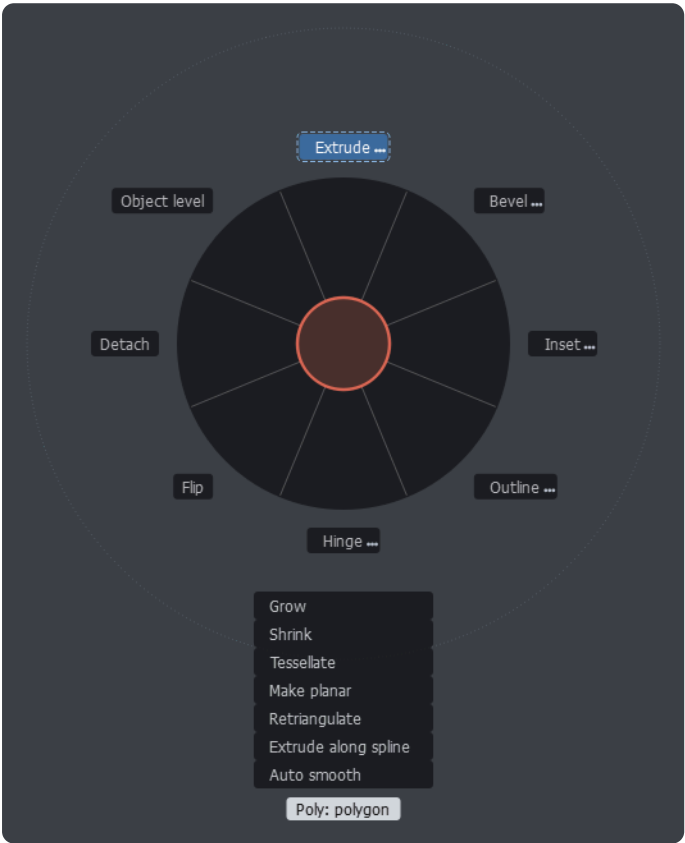
2. The dial of the object under the cursor

The dial looks at the object **under the cursor** first, and at the selection only when the cursor is over empty space. Point at a spline and open Alt+1: you get spline tools even while a box is selected - and the command runs on that spline, which MarkingForge selects in the same undo step. Pointing at one of

several selected objects keeps the whole selection. The modifier stack dial (Alt+6) shows the stack of the object you point at the same way. A flick reads the object where you **PRESSED** the key, so it works the same even when the movement ends over empty space. The hotbox and type-to-search run 3ds Max commands and never change the selection.

3. The suggested next command

MarkingForge notices pairs - Inset followed by Extrude, say. Once a pair has repeated twice, the command you usually run next is the dial's **outlined tile** when it opens, and a **tap** of the key - press and release without moving - runs it. That works when the command sits on that dial and the dial has no tap item of its own (an explicit tap item always wins). It is only a suggestion: aiming anywhere else works as always, and nothing on the dial moves. The memory lasts for the session.



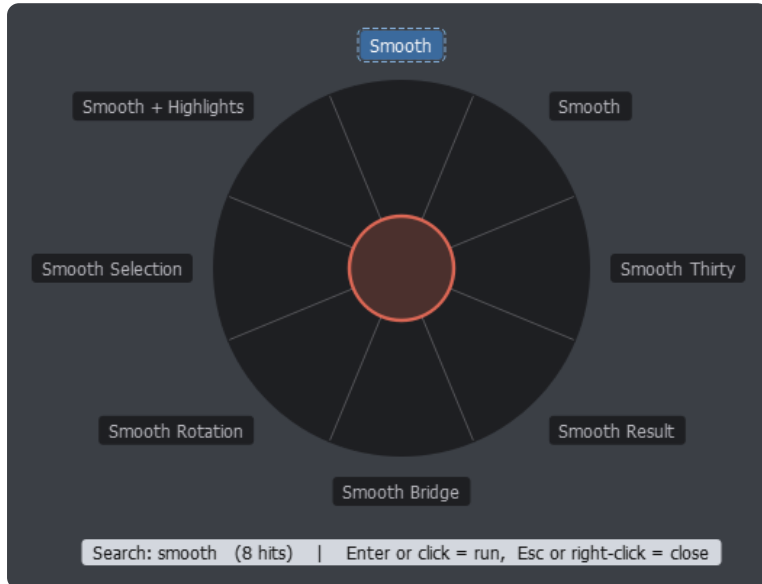
After Inset, Extrude is the outlined tile - a tap runs it.

4. Hints you can see

On the dial	Means
Dotted circle outside the ring	The settings rim (tutorial 4) - brighter once you are past it.
Dashed circle round the centre	A queue is waiting: come back inside it to add the command (trick 1).
Dashed outline of a direction	That direction is empty - releasing there cancels.
> after a caption	A submenu: move through it without releasing and it opens where the cursor is (tutorials 1 and 13).
The circle in the centre	Release inside it to cancel - always. Since 1.0.9 it is twice as large (36 px), easy to hit without looking.
Outlined tile	What a tap runs - the suggested next command (trick 3); Enter too when the dial is pinned (trick 5).

5. Search by typing (switch it on first)

- 1 Switch on **MarkingForge > Experimental features > Type-to-search**.
- 2 Hold a dial key and **type letters** - for example `smooth`. The dial turns into a search over every 3ds Max command and **pins itself**, so you can let go of the key.
- 3 Aim at a result, or just press **Enter** for the best match. **Backspace** removes a letter, **Esc** closes the search.



Alt+1, then `smooth`: commands with that name, the best match outlined.

Typing works with the dials on **Alt** and **Shift+Alt**. With **Ctrl** held, letters go to 3ds Max as shortcuts (Ctrl+S, Ctrl+Z...) on purpose - so there is no search on the Ctrl+Alt dials.

The key map

Digits 1-8 with **Alt**, **Ctrl+Alt** and **Shift+Alt**. The digit is the place, the modifier the bank - count from a key you know.

Key	Dial	Key	Dial	Key	Dial
Alt+1	Modelling	Ctrl+Alt+1	Transform	Shift+Alt+1	Materials
Alt+2	Viewport shading	Ctrl+Alt+2	Modifiers	Shift+Alt+2	Animation
Alt+3	Create	Ctrl+Alt+3	Show and hide	Shift+Alt+3	Scene
Alt+4	View	Ctrl+Alt+4	Viewports	Shift+Alt+4	File
Alt+5	Selection	Ctrl+Alt+5	Viewport display	Shift+Alt+5	Selection sets
Alt+6	Modifier stack	Ctrl+Alt+6	Snaps and grid	Shift+Alt+6	UV mapping
Alt+7	Undo	Ctrl+Alt+7	Align and pivot	Shift+Alt+7	Tools and setup
Alt+8	Recent commands	Ctrl+Alt+8	Render	Shift+Alt+8	Lights and cameras

Print the cheat sheet and keep it beside the keyboard for the first week: **MarkingForge > Dial cheat sheet (HTML + PDF)...** - your saved dials as 3x3 squares with their keys and mouse buttons, sorted by key.

MarkingForge - dial cheat sheet

2026-10-03 06:19 - 24 dials - sorted by shortcut, because that is what the hand presses

Alt+1 Modelling

slot 1

1. when window under the cursor matches slate

⌘ NI Pick from selected	↑ N New material	↩ NE Edit selected's material
⇧ W Get from object	Materials (Slate)	→ E Assign to selection
⌘ SW Material Explorer	↓ S Material/Map Browser	↘ SE Show in viewport

2. when window under the cursor matches trackView

⌘ NI Curve mode	↑ N Tangents: auto	↩ NE Tangents: smooth
⇧ W Isolate curve	Track View	→ E Tangents: linear
⌘ SW Frame all	↓ S Flatten	↘ SE Tangents: stepped

3. when window under the cursor matches particleView

⌘ NI Rename	↑ N Move up	↩ NE Copy
⇧ W Move left	Particle View	→ E Move right
⌘ SW Cut	↓ S Move down	↘ SE Paste

4. when window under the cursor matches sceneExplorer

⌘ NI Zoom to selected	↑ N Select all	↩ NE Select children
⇧ W Unhide all	Scene Explorer	→ E Hide
⌘ SW Unfreeze all	↓ S Isolate	↘ SE Freeze

List below the dial

1. Expand all
2. Collapse all
3. Find selected
4. Rename

5. when window under the cursor matches commandPanel

Modifier stack. Eight modifiers FROM THE TOP of the selected object's stack. Requires EXACTLY ONE selected object - otherwise the dial opens empty and says so in its footer.

6. when modifier = Unwrap_UVW

⌘ NI Stitch	↑ N Quick relax	↩ NE Pack
⇧ W Weld selected	UV (Unwrap)	→ E Edge loop
⌘ SW Break	↓ S Straighten	↘ SE Edge ring

List below the dial

1. Open UV Editor

7. when modifier = Edit_Poly and subobject: 1 - usually vertex

⌘ NI Object level	↑ N Chamfer	↩ NE Weld
⇧ W Target Weld	Edit Poly: vertex	→ E Connect
⌘ SW Remove	↓ S Extrude	↘ SE Break

List below the dial

1. Grow
2. Shrink
3. Remove isolated

8. when modifier = Edit_Poly and subobject: 2 - usually edge

⌘ NI Object level	↑ N Connect	↩ NE Chamfer
⇧ W Remove	Edit Poly: edge	→ E Loop
⌘ SW Weld	↓ S Extrude	↘ SE Ring

List below the dial

1. Split
2. Turn
3. Grow
4. Shrink

9. when modifier = Edit_Poly and subobject: 3 - usually border / spline

⌘ NI Object level	↑ N Cap	↩ NE Chamfer
⇧ W Grow	Edit Poly: border	→ E Extrude
⌘ SW Shrink	↓ S Create shape	↘ SE Connect

List below the dial

1. Bridge

The printable cheat sheet.

PART 2

Making it yours

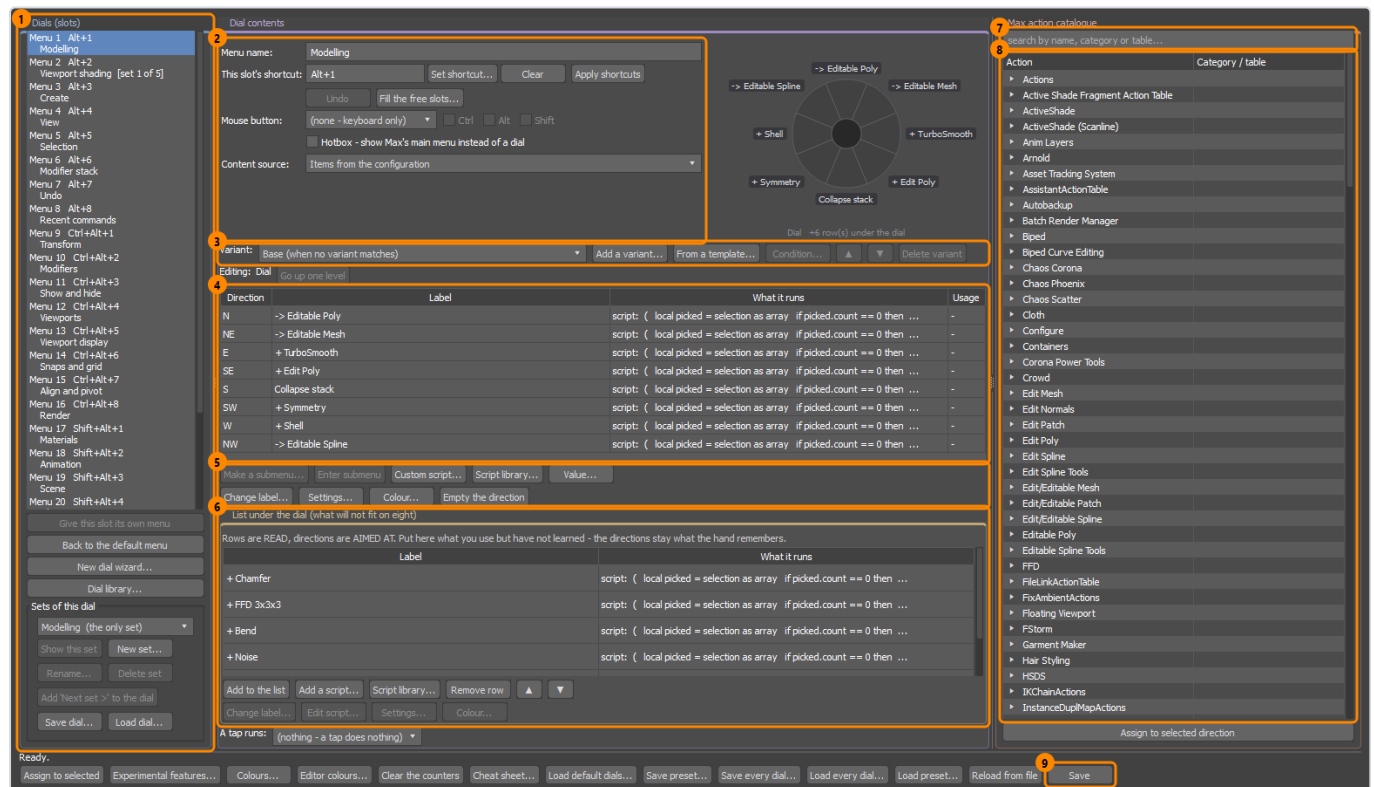
Sixteen tutorials in the menu editor. Nothing you do here reaches the dials until you press **Save** - experiment freely, and use **Reload from file** to throw changes away. If the file changed on disk while the editor was open, Save offers a separate copy instead of overwriting it. The one exception is the editor's own colours (tutorial 17): they are not part of your dials and are kept as soon as you press OK.

7

A tour of the editor

2 min · You will learn: where everything is

Open **MarkingForge > Menu editor...**



The menu editor.

- 1. Dials (slots)** - the 24 slots and their keys. Pick the dial to edit; right-click it to save or load it.
- 2. Dial contents** - name, shortcut, mouse button, hotbox, where its contents come from.
- 3. Variant** - the dial's context versions and their conditions.
- 4. Directions** - N ... NW: label, what runs, how often you used it.
- 5. Direction tools** - submenu, script, script library, value, label, settings, colour, empty.
- 6. List under the dial** - extra rows.

7. **Search** - find any 3ds Max command by name.
8. **Catalogue** - all 4000+ actions and macroscripts, by category.

9. **Save** - writes your dials; the plugin reloads in a fraction of a second.

Beside the dial contents the editor **draws the dial** you are editing - click a wedge to select its direction. Under the slot list: **New dial wizard...**, **Dial library...** and **Sets of this dial**; a dial with more than one set shows them as **tabs** above its contents (tutorials 20-22). The bottom bar holds what concerns all dials: **Colours...**, **Editor colours...**, the cheat sheet, presets, **Save every dial...** and **Load every dial...** (tutorials 16-18).

Closing the editor never loses work silently. With unsaved changes, Esc or the window's X asks: **Save**, **Discard** or **Cancel**. Shortcuts you set but did not apply are offered for applying first.

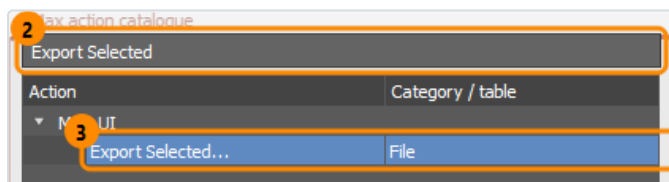
8

Put a command on a direction

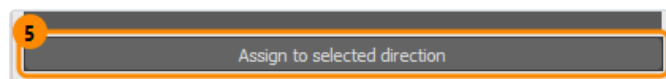
3 min · You will learn: the one skill everything else builds on

Example: on the **File** dial (Shift+Alt+4) replace *Save copy as* (south-east) with **Export Selected**.

- 1 In the dial list pick **Menu 20** (Shift+Alt+4, File).
- 2 Type **Export Selected** into the search field.
- 3 Click **Export Selected...** in the catalogue.
- 4 Click the **SE** row in the directions table.
- 5 Press **Assign to selected direction** - or double-click the command, or drag it onto the row.
- 6 Press **Save**.



Steps 2 and 3: type, then click the command.



Step 5: the button under the catalogue.

Direction	Label	What it runs	Usage
N	Save	action: Main UI	-
NE	Save as...	action: Main UI	-
	Save incremental	action: Main UI	-
SE	Save copy as...	action: Main UI	-
S	Open...	action: Main UI	-
SW	Import...	action: Main UI	-
W	Merge...	action: Main UI	-
NW	Hold	action: Main UI	-

Step 4: the SE row selected - it still holds Save copy as.

Direction	Label	What it runs	Usage
N	Save	action: Main UI	-
NE	Save as...	action: Main UI	-
E	Save incremental	action: Main UI	-
SE	Export Selected...	action: Main UI (name from Max)	-
S	Open...	action: Main UI	-
SW	Import...	action: Main UI	-
W	Merge...	action: Main UI	-
NW	Hold	action: Main UI	-

After step 5: SE runs Export Selected. (*name from Max*) means the caption is 3ds Max's own - tutorial 9 shortens it.

TRY IT

Hold Shift+Alt+4 in the viewport - south-east is Export Selected.

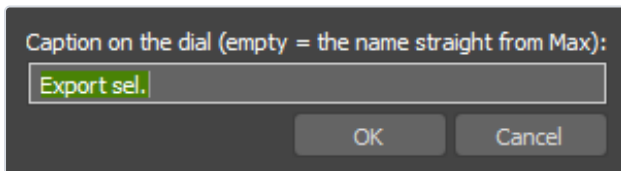
The same command appears in several categories (Select All exists in seven). The *Category / table* column tells them apart - for the scene commands pick *Main UI*.

9

A shorter label

1 min · You will learn: to keep captions readable

- 1 Select the direction.
- 2 Press **Change label...**
- 3 Type the caption and press OK. Empty = 3ds Max's own name.
- 4 Save.



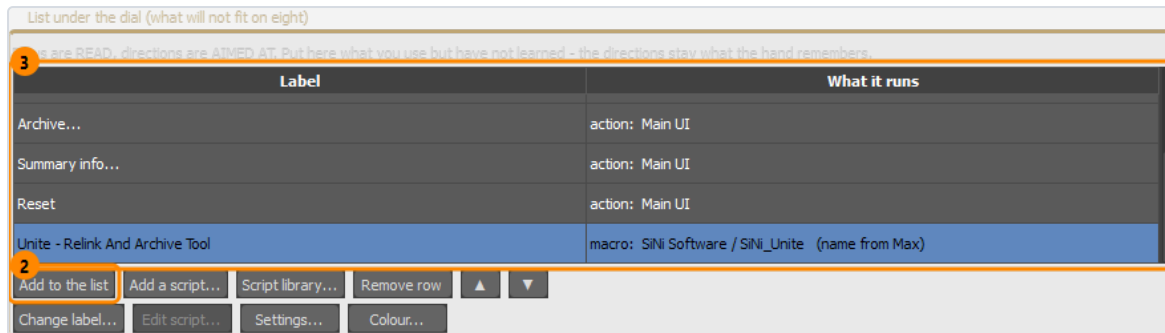
Two words beat a cut-off name on the dial.

10

Rows under the dial

2 min · You will learn: to keep more than eight commands on one key

- 1 Pick the dial.
- 2 Find the command in the catalogue.
- 3 Press **Add to the list** (or drag the command into the list).
- 4 Reorder with ▲ ▼, remove with **Remove row**. Up to twelve rows.
- 5 To edit a row, select it: **Change label...**, **Edit script...**, **Settings...** and **Colour...** sit in the list's own bar. **Add a script...** adds a row that runs your own MAXScript.
- 6 Save.



The list pane: the rows, and Add to the list under them.

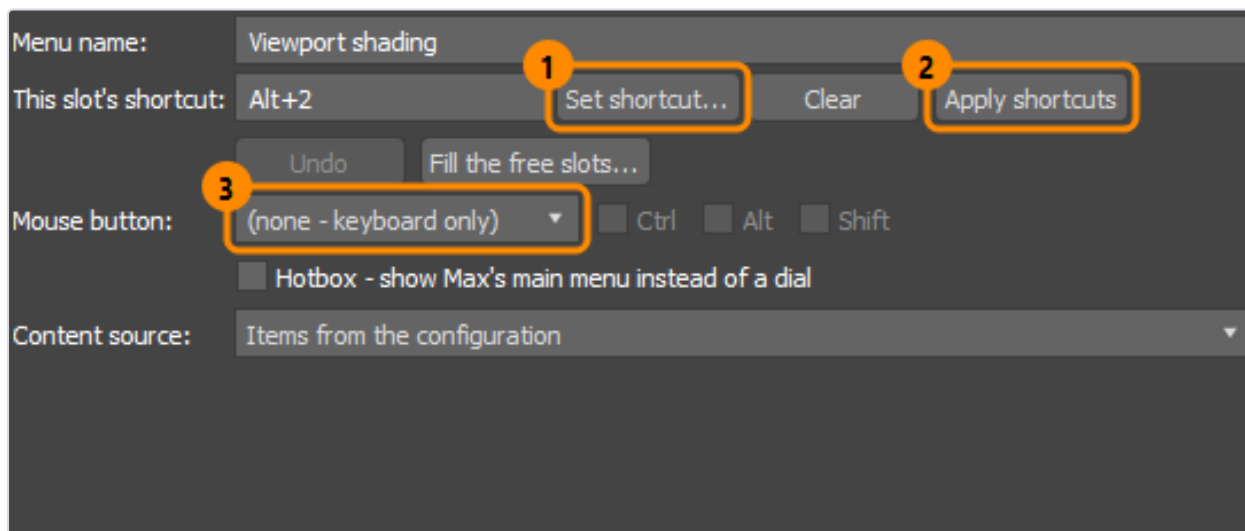
Put here what you use but have not learned. Promote a row to a direction once your hand wants it.

11

Change a dial's key, or open it with the mouse

2 min · You will learn: shortcuts and mouse buttons

- 1 Pick the dial.
- 2 Click **Set shortcut....** A small window opens: press the new keys there (Shift combinations work too) and click **Set**. While it is open the dials do not open, so a key that already opens a dial can be recorded - it moves from that dial to this one. **Clear** (or Backspace) removes the key.
- 3 Press **Apply shortcuts** - the keys go into 3ds Max; your other shortcuts are kept and backed up. **Undo** beside it puts back the shortcuts from before; **Fill the free slots...** offers the standard key to every slot without one.
- 4 Optional: choose a **Mouse button** and tick at least one of Ctrl / Alt / Shift - for example Ctrl+Shift + *middle button (wheel)* for the hotbox.
- 5 Save.



The dial settings: Set shortcut (1), Apply (2), mouse button (3).

The keys are also in 3ds Max's **Customize > Hotkey Editor** as *MarkingForge: Menu 1 ... 24*.

12

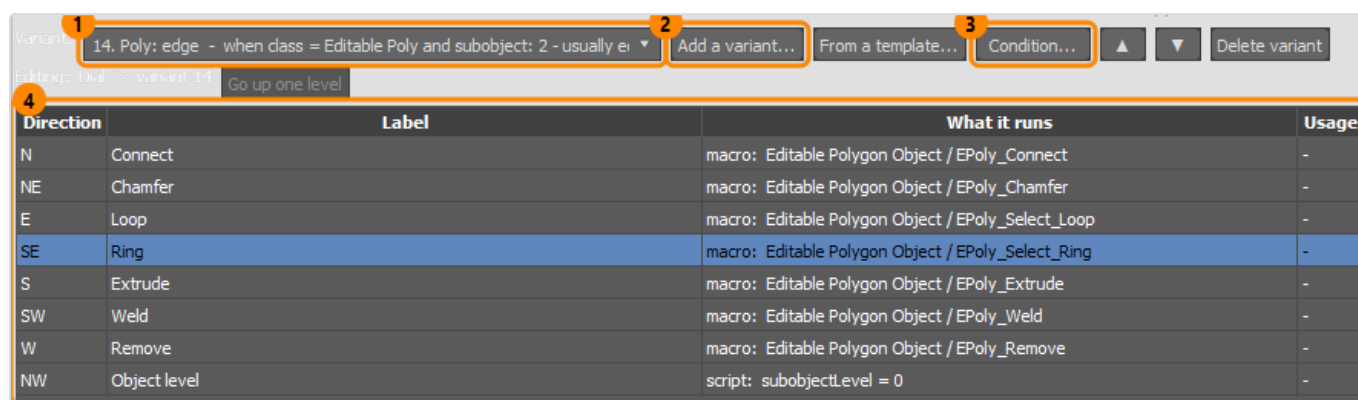
A context variant

4 min · You will learn: to make one key do different things in different situations

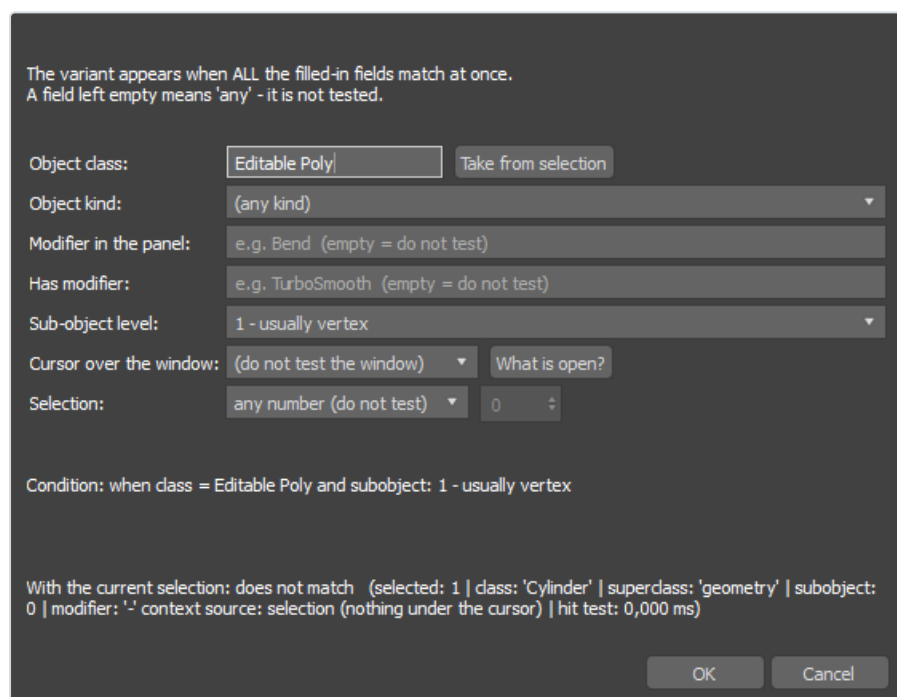
Example: give a dial its own contents when you are on the edges of an Editable Poly.

- 1 Pick the dial and press **Add a variant...** - it starts as a copy of what you see.

- 2 In the condition window select an Editable Poly in 3ds Max, go to edge level and press **Take from selection:** class and sub-object level fill in.
- 3 OK. The **Variant** list now shows your variant - fill its directions like any dial.
- 4 Order matters: the first matching variant wins. Use ▲ ▼ to keep specific ones above general ones.
- 5 Save.



The Modelling dial's built-in edge variant: the variant list (1), Add a variant (2), Condition (3) and the variant's own directions (4).



The condition window (here: Editable Poly, vertex level). Empty fields are not tested.

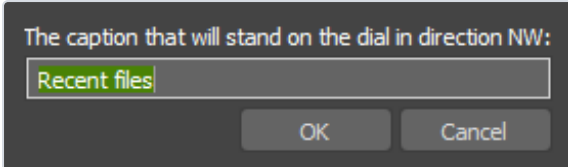
Conditions can also ask about the **window under the cursor** (Slate, Track View, Scene Explorer...), the modifier open in the panel, the number of selected objects, an object name pattern or a layer.

13

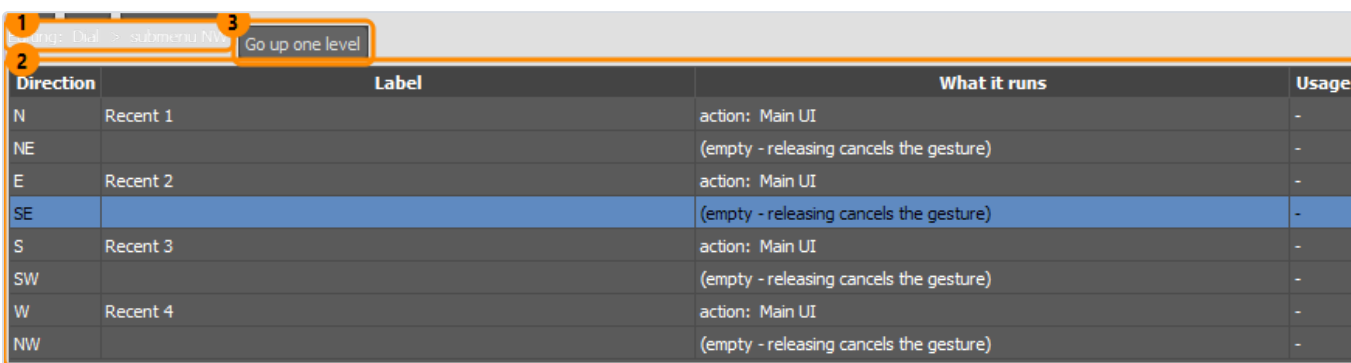
A submenu

3 min - You will learn: to put a whole dial behind one direction

- 1 Select a direction and press **Make a submenu...**
- 2 Type its caption - it stands on the dial.
- 3 You are now inside the submenu (the path says *Editing: Dial > submenu NW*). Fill its directions.
- 4 Press **Go up one level** to return. Save.



Step 2: the caption for the submenu.



Step 3: inside the submenu - the path (1), its own directions (2), and the way back (3).

TRY IT

On the dial, move through the submenu's direction without releasing - the submenu opens in place, the dial does not move; come back inside the dial to pick, into its centre to go back up.

TRY IT

On the dial, move through the submenu's direction without releasing - the submenu opens in place, the dial does not move; come back inside the dial to pick, into its centre to go back up.

Submenus in the built-in dials. Open them to see what a submenu is good for - groups that would not fit on eight directions:

Dial	Direction	Submenu
Alt+3 Create	S, SW, W, NW	Shapes, More primitives, Helpers, Cameras and lights
Ctrl+Alt+2 Modifiers	SW, NW	Deform (Bend, Twist, Taper, FFD...), Geometry (Mirror, Array, Cap Holes, OpenSubdiv...)
Alt+1 Modelling, nothing selected	SW, W	Shapes, More primitives

In the editor, the drawing of the dial beside the form marks a submenu with >; **double-click** it there (or its row in the table) to step inside.

Scripts on a dial - ready-made or your own

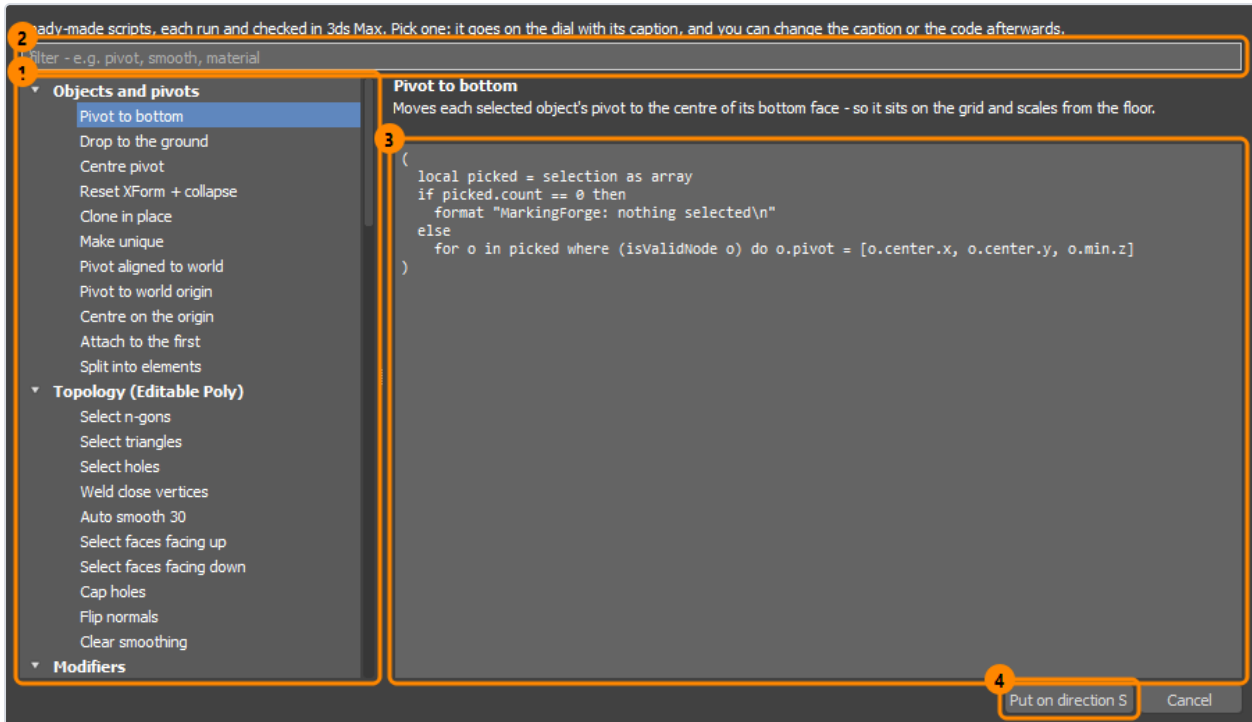
6 min · You will learn: to run a script from a gesture, with or without writing one

A direction or a list row can run MAXScript. You do not have to write it: the **Script library** holds 102 ready-made scripts in eleven groups - objects and pivots, topology, modifiers, advanced modelling (loops and rings, chamfer, bridge, bevel, planarize, symmetrize, greeble...), UV mapping, materials and display, selection and transforms, layout and placement, shapes and splines, scene and layers, cameras, lights and render - each run in 3ds Max on a test scene, and with nothing selected, before it went into the list.

A. From the library - no code

- 1 Pick the dial and select a direction - or, to keep your directions, use the list under the dial.
- 2 Press **Script library...** (under the directions, or in the list's bar).

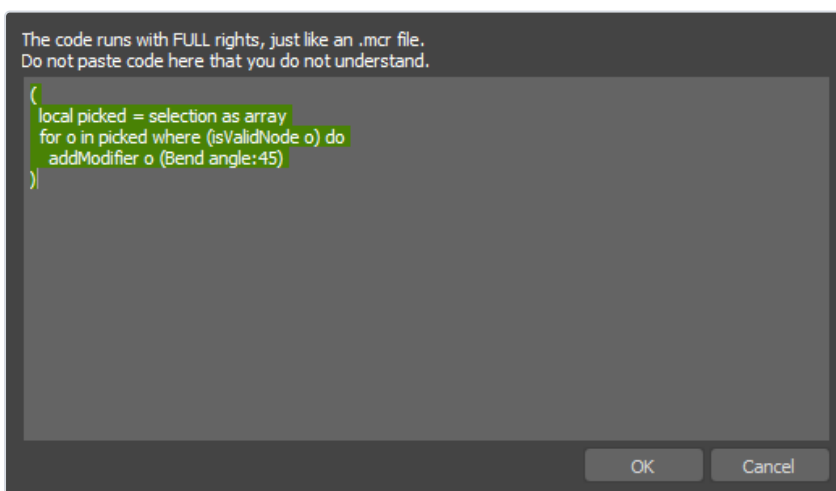
- 3 Click a script on the left: the right side says what it does and shows its code. Type in the filter to narrow the list - *pivot, smooth, material*.
- 4 Press **Put on direction S** - the button names the direction you selected - (or **Add to the list**), then **Save**.



The script library - groups on the left, what the script does and its code on the right.

B. Your own code

- 1 Select a direction and press **Custom script...** - it reads **Edit script...** once the direction holds a script.
- 2 Paste or type the code and press OK.
- 3 Give it a caption with **Change label...**
- 4 Save.



A script that adds a Bend to every selected object.

Four scripts every modeller uses

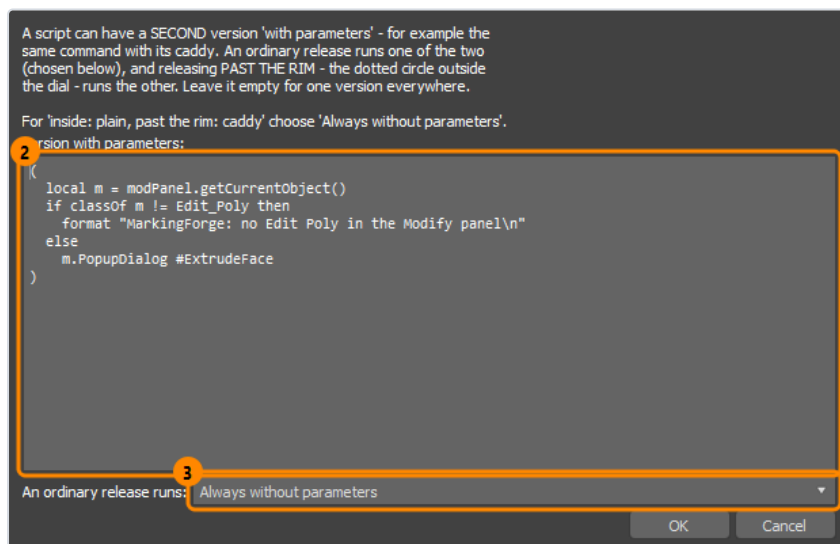
All four are in the library. Put them on the list under the Modelling dial (**Menu 1**, Alt+1) and they are one gesture away whatever you model.

Script	Try it like this
Pivot to bottom	Create a sphere, hold Alt+1, pick the row: the pivot jumps to the bottom of the sphere. Pair it with Drop to the ground and objects stand on the grid.
Select n-gons	Convert a cylinder to Editable Poly, pick the row: both caps light up at polygon level, and the MAXScript Listener says how many were found. Fix them before smoothing.
Reset XForm + collapse	Scale a box to 300 % and rotate it, then pick the row: the scale and rotation go into the mesh and the transform is clean - what Shell, Symmetry and exporters expect.
Smoothing on/off	Give a few objects TurboSmooth, select them all, pick the row: every TurboSmooth switches off at once; pick it again and they come back.

- 1 Select **Menu 1** in the editor's left pane.
- 2 In the list bar press **Script library...**, pick **Pivot to bottom**, press **Add to the list**.
- 3 Repeat for the other three.
- 4 Save, then hold Alt+1: the four rows sit under the dial.

C. Two versions: plain inside, the caddy past the rim

- 1 Select a direction that runs a script and press **Settings...**
- 2 Paste the second version into **Version with parameters** - for Extrude on an Edit Poly modifier:
`(modPanel.getCurrentObject()).PopupMenu #ExtrudeFace`
- 3 Choose **Always without parameters**: an ordinary release runs the plain code, releasing past the dotted rim runs the caddy version.
- 4 OK, then Save. In the table the script now reads *[+ with parameters]*.

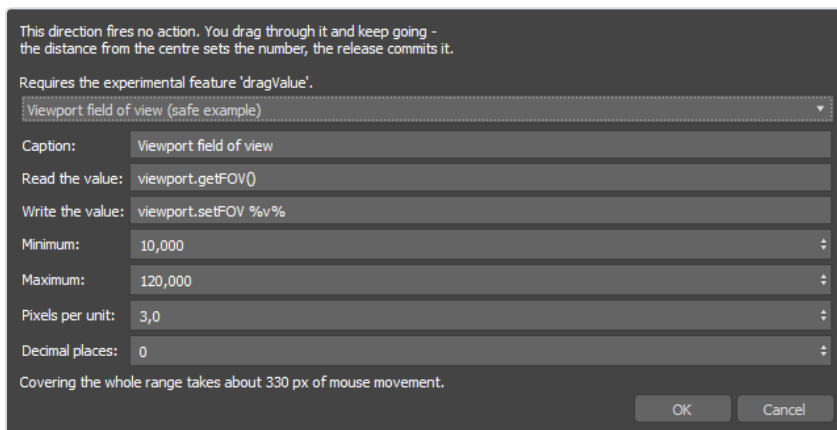


Settings... on a script: the version with parameters and which one an ordinary release runs.

The library's **Extrude (Edit Poly)** comes set up this way. Scripts run with full rights: only use code you understand. The Reference (chapter 8, *Item types*) lists every library script, prints the code of all four examples and explains the shape a script should have.

A **value** direction runs no command. You hold the key, move into that direction and keep going: the **distance** from the centre sets a number, shown next to the caption while you move, and the **release** writes it into the scene. Two lines of MAXScript define it - how to **read** the number when the cursor enters, and how to **write** it on release, where `%v%` stands for the number.

- 1 Switch on **MarkingForge > Experimental features > Direction that sets a value** (in the editor: **Experimental features...** in the bottom bar). Without it the direction does nothing and the dial says so.
- 2 In the editor pick a dial, select an **empty direction** and press **Value...**
- 3 Pick one of the examples below from the list at the top - it fills every field - or type the fields yourself from the tables.
- 4 Read the line at the bottom of the window: it says how far the mouse must travel to cover the whole range (range × pixels per unit). Between about 150 and 500 pixels is comfortable.
- 5 **OK**, then **Save**. The dial in 3ds Max picks the change up at once.



The value window with the first example, the viewport field of view.

Example 1 - Viewport field of view

Zoom the lens of the perspective view without touching the camera - the safe one: needs no selection and changes only the view.

Field	Value
Caption	Viewport field of view
Read the value	<code>viewport.getFOV()</code>
Write the value	<code>viewport.setFOV %v%</code>
Minimum / Maximum	10 / 120
Pixels per unit	3
Decimal places	0

TRY IT

In a perspective view drag into the direction and on: the view widens as you move out, narrows as you come back; release at the angle you like.

Example 2 - Time slider frame

Scrub the animation from the dial - a frame per three pixels.

Field	Value
Caption	Time slider frame
Read the value	<code>sliderTime.frame</code>
Write the value	<code>sliderTime = %v%</code>
Minimum / Maximum	0 / 100
Pixels per unit	3
Decimal places	0

TRY IT

Move out slowly and watch the time slider follow; release on the frame you want.

Example 3 - TurboSmooth iterations of the selection

Raise or lower the smoothing of every selected object that has TurboSmooth - 40 pixels per step, so each step is easy to hit.

Field	Value
Caption	TurboSmooth iterations of the selection
Read the value	<code>selection[1].modifiers[#TurboSmooth].iterations</code>
Write the value	<code>for o in selection where o.modifiers[#TurboSmooth] != undefined do o.modifiers[#TurboSmooth].iterations = %v%</code>
Minimum / Maximum	0 / 4
Pixels per unit	40
Decimal places	0

TRY IT

Select a few objects with TurboSmooth; the number next to the caption steps 0, 1, 2, 3, 4 as you move.

Example 4 - Uniform scale of the selection

Scale every selected object evenly, from a tenth to five times, with two decimals.

Field	Value
Caption	Uniform scale of the selection
Read the value	<code>selection[1].scale.x</code>
Write the value	<code>for o in selection do o.scale = [%v%, %v%, %v%]</code>
Minimum / Maximum	0.1 / 5
Pixels per unit	60
Decimal places	2

TRY IT

Select objects, move out: they grow in the viewport only after the release - the number shows where you are while you move.

Example 5 - Thickness of the selected splines

Make splines renderable and set their thickness - a quick tube from a line.

Field	Value
Caption	Thickness of the selected splines
Read the value	<code>selection[1].render_thickness</code>
Write the value	<code>for o in selection where superClassOf o == shape do (o.render_renderable = true; o.render_displayRenderMesh = true; o.render_thickness = %v%)</code>
Minimum / Maximum	0.1 / 20
Pixels per unit	15
Decimal places	1

TRY IT

Draw a few lines or circles, select them and set the thickness in one move.

? next to the caption means the read gave no number - usually nothing (or the wrong kind of object) is selected. The release then writes nothing. Examples 3-5 act on **every** selected object and read the first one; they skip objects the code does not fit (no TurboSmooth, not a shape).

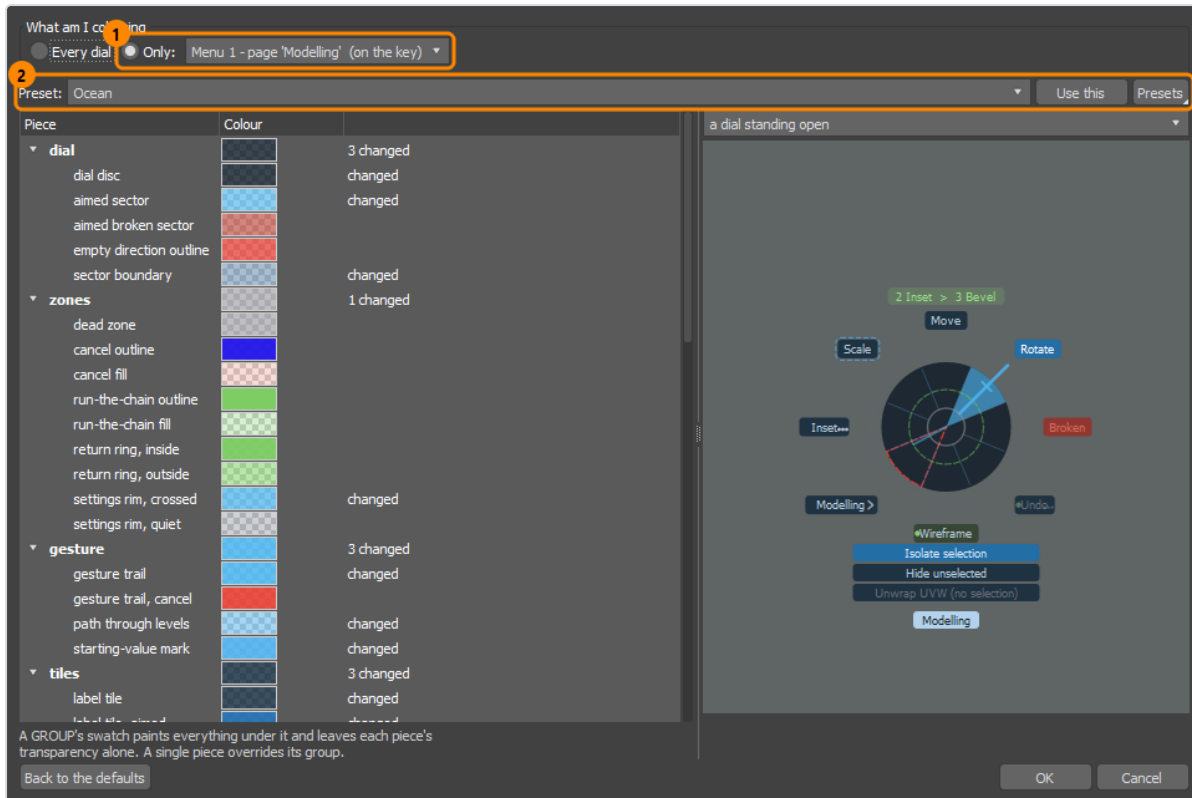
Moving back towards the centre lowers the number again; returning into the centre itself gives up - nothing is written.

16

Colours and ready-made looks

3 min · You will learn: to recognise a dial before reading it

- 1 Press **Colours...** in the bottom bar of the editor.
- 2 Under **What am I colouring** choose **Every dial**, or **Only** this one. For a dial with sets the **Only**: list also offers *Menu N - every page* and each page on its own (*Menu N - page 'UV'*) - the pages the mouse wheel turns (tutorial 20).
- 3 Pick a look from **Preset:** and press **Use this**. There are thirty ready-made looks - Graphite, Ocean, Ember, Neon, High contrast and more, five of them with light tiles - and the *MarkingForge default*.
- 4 Or click a group swatch (dial, zones, tiles, text, list...) or a single piece. The list above the preview shows the moment you are colouring: *a dial standing open, aiming at a direction, releasing would cancel, a chain is waiting*.
- 5 OK, then Save.



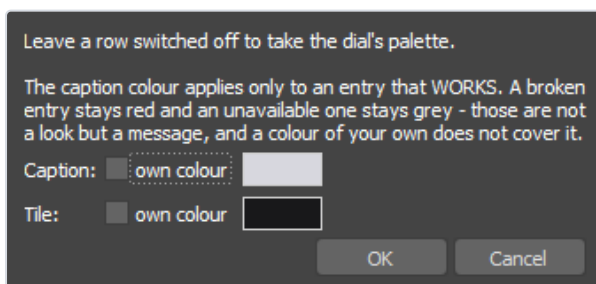
Colours...: what you colour - here one page of Menu 1 (1) - and the preset row (2).

Your own looks. Presets > **Save these colours as a preset...** keeps what you made under a name; **Export this preset to a file...** writes one look to a `.mfcolors` file for another computer, and **Import a preset from a file...** brings it in. A dial that shows the default menu has no colours of its own: press **Give this slot its own menu** first.



The thirty ready-made looks.

For one command only - a red caption on a destructive one, say - select its direction and press **Colour...**:



One item's caption and tile colour.

17

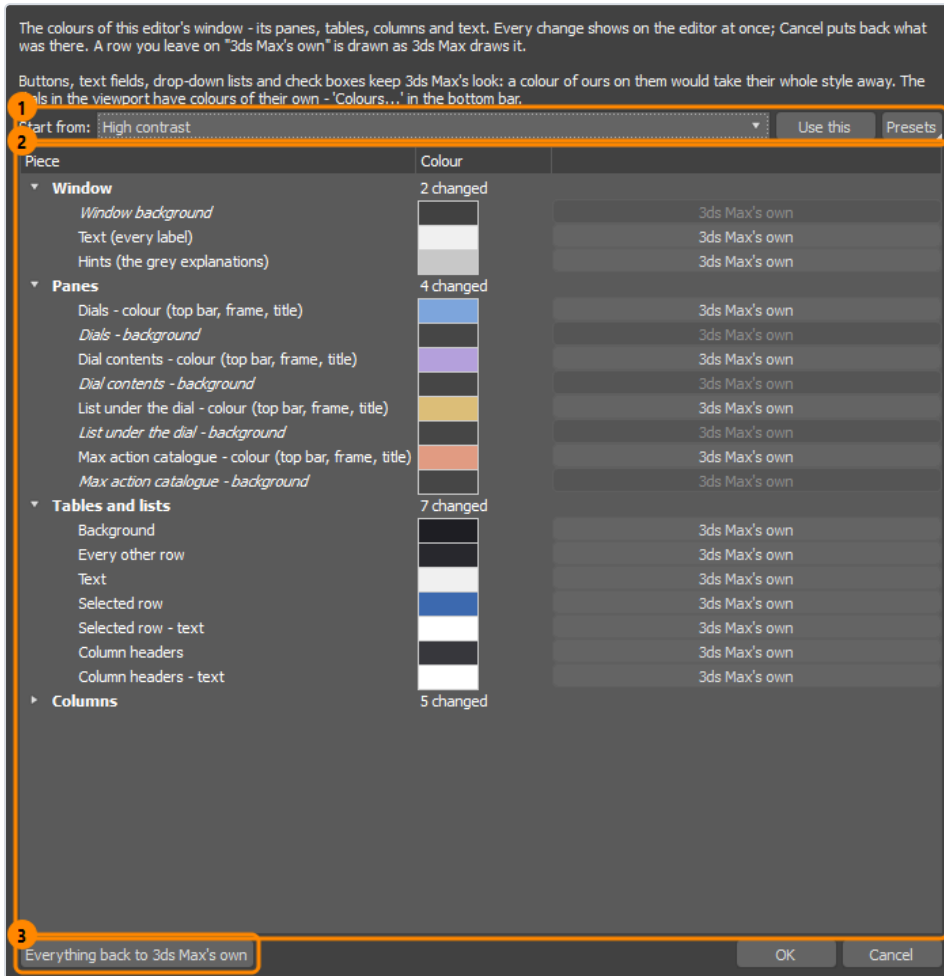
The editor's own colours

2 min - You will learn: to colour the editor's window - not the dials

1

Press **Editor colours...** in the bottom bar.

- 2 Pick a starting point in **Start from:** and press **Use this** - 21 ready-made looks (High contrast, Calm grey, Nord, Espresso...) besides *3ds Max's own*.
- 3 Change any piece: the window, each pane, the tables and lists, every column on its own. The editor changes as you pick. **3ds Max's own** on a row drops that colour.
- 4 OK keeps it - no Save needed. Cancel puts back what was there.



Editor colours...: the starting points (1), every piece of the window (2), everything back to 3ds Max's look (3).

Buttons, text fields, drop-down lists and check boxes keep 3ds Max's look. The colours are kept in `editor_colors.json` beside your dials and are not part of them; **Presets** saves, exports and imports your own looks, as in tutorial 16.

18

Print, share, reset - and add commands to a quad menu

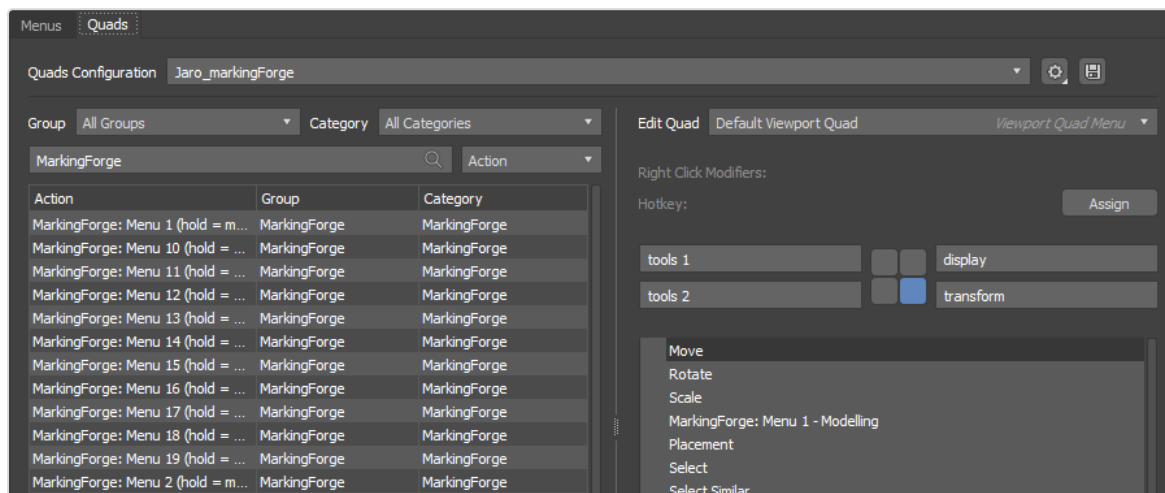
3 min · You will learn: the bottom row and 3ds Max's own menus



Cheat sheet (1), Load default dials (2), Save preset (3), Load preset (4), Save (5), Save every dial (6), Load every dial (7).

- 1 **Cheat sheet...** - a printable PDF of your saved dials, with keys and mouse buttons. Save first: unsaved edits are left out.
- 2 **Save preset...** - your layout in one file to share. **Load preset...** shows what a file holds before loading and removes scripts from other people's presets by default.
- 3 **Load default dials...** - the built-in layout back; by default only dials without their own menu.

- 4 **Save every dial...** - each of the 24 dials, and every other set of a dial, as a `.mfdial` file of its own in a new dated folder. **Load every dial...** puts a whole such folder back: what each dial showed, its sets in their order and the colours of every dial. Scripts from the files are removed unless you keep them, dials the folder does not hold stay as they are, and nothing is written until Save.
- 5 Any MarkingForge command can go into a right-click quad menu: **Customize > Menu Editor > Quads**, search `MarkingForge`, drag an action into a quad, **Done**.

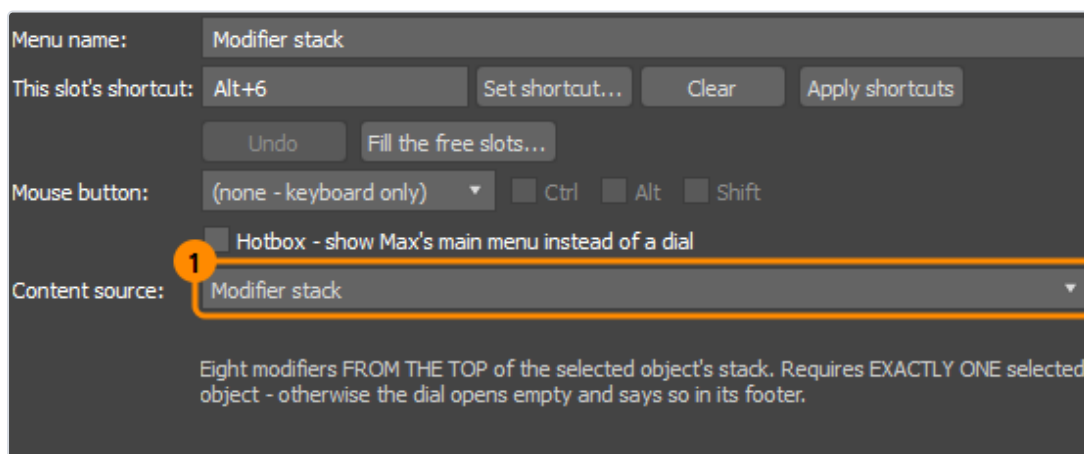


Menu Editor, Quads tab, filtered to MarkingForge.

19 What a dial shows - content sources

6 min · You will learn: dials that fill themselves: the modifier stack, undo, recent commands, selection sets

Content source, in the *Dial contents* pane, decides where a dial's eight directions come from. *Items from the configuration* is an ordinary dial - the commands you put on it. Every other choice is a **live** dial: it fills itself each time it opens, from the scene or from what you did, and the eight directions you typed are kept aside (they come back if you switch the source back).



Content source, under the shortcut and the mouse button.

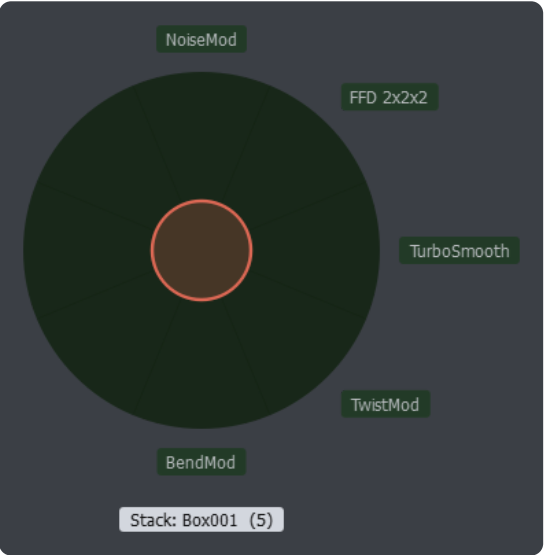
Choice	The directions are	Releasing on one	Built-in dial
Items from the configuration	what you put there	runs it	most dials
Modifier stack	the top eight modifiers of the ONE selected object	makes it the current modifier in the Modify panel	Alt+6
Undo stack	your last eight operations, by name	undoes back to it - north one step, north-east two, and so on	Alt+7

Recent commands	the eight commands you ran last, newest first (from any dial, a tap or the search)	runs it again	Alt+8
Selection sets	the scene's named selection sets	selects that set	Shift+Alt+5

Undo stack, Recent commands and Selection sets each have a switch in **Experimental features**; all three are on out of the box. The **Hotbox** tick box above Content source is the sixth way to fill a dial: every menu of 3ds Max's menu bar as tiles (Alt+2).

Modifier stack - step by step

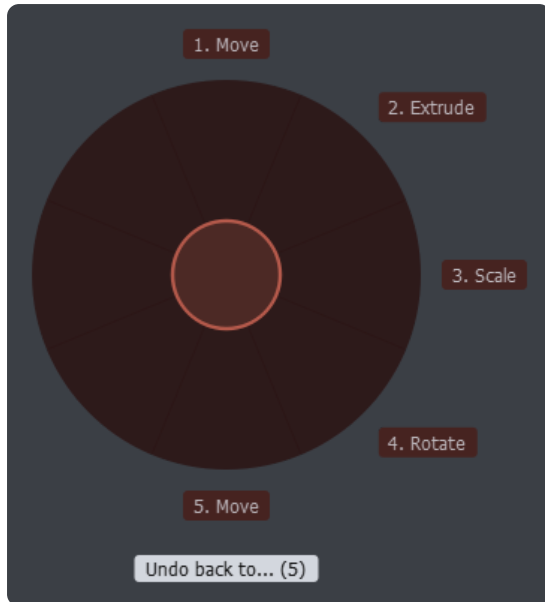
- 1 Pick a dial you want to turn into a stack dial (or give a free slot its own menu).
- 2 Set **Content source** to **Modifier stack** and press **Save**.
- 3 In the scene select ONE object with a few modifiers - a box with Bend, Twist and TurboSmooth.
- 4 Hold the dial's key: the modifiers sit on the directions, the top of the stack first. Release on Bend - the Modify panel jumps to Bend.



Alt+6 - the stack of the selected box.

Undo stack - step by step

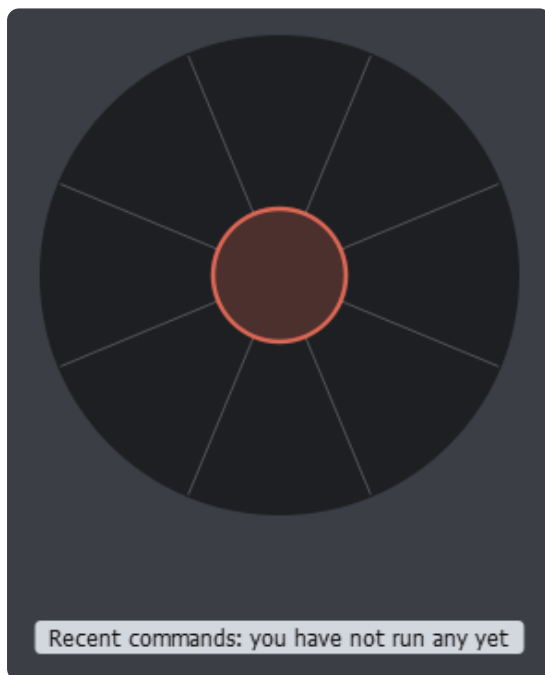
- 1 On a dial set **Content source** to **Undo stack** and Save.
- 2 Do a few things: move, extrude, add a modifier.
- 3 Hold the key: your operations are listed by name, the last one north. Release three directions round - three steps undone in one move.



Alt+7 - undo by name.

Recent commands - step by step

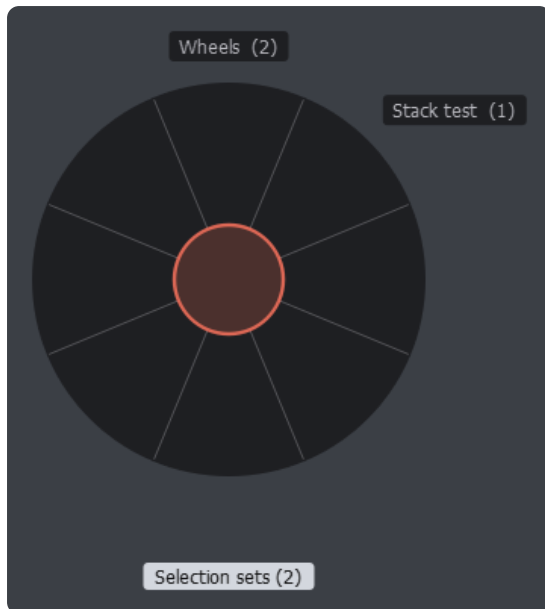
- 1 Set **Content source** to **Recent commands** and Save.
- 2 Run a few commands from any dial (a flick, a tap, a list row or the search).
- 3 Hold the key: they are there, newest north. A flick up repeats the last one. The history starts afresh each time 3ds Max starts.



Alt+8 - recent commands.

Selection sets - step by step

- 1 Select a few objects and name a selection set in the main toolbar's **Named Selection Sets** field (type a name, press Enter). Make two or three sets.
- 2 Set **Content source** to **Selection sets** and Save.
- 3 Hold the key: each set is a direction with its object count. Release on one - it is selected.



Shift+Alt+5 - the scene's selection sets.

A live source on one variant only

- 1 On a dial press **Add a variant...**; in the condition window that opens set **Cursor over the window** to **Command panel** and press OK.
- 2 With the new variant selected, set **Content source** to **Modifier stack**.
- 3 Save. The same key now shows your commands over a viewport and the modifier stack over the command panel - the built-in Modelling dial does exactly this.

A live dial with nothing to show opens empty and says why in its footer - *Modifier stack: select ONE object, Undo: nothing to undo, Selection sets: the scene has none*. That is not an error.

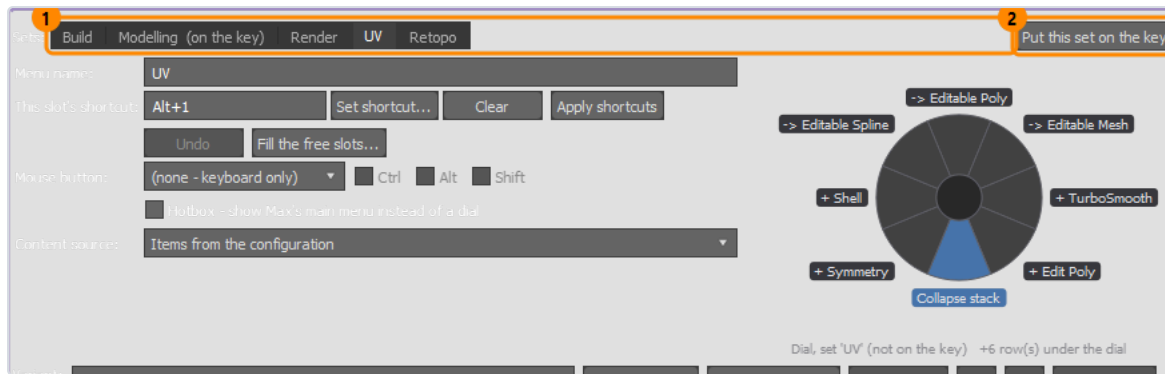
20

Pages - several dials on one key, turned with the mouse wheel

4 min - You will learn: to keep Modelling, UV and Retopo on one key and flip between them while you hold it

A dial can keep several **sets** of contents under names. The key opens one of them - the set *on the key*; the others are the dial's next **pages**. Everything else stays: the key, the mouse button, the other 23 dials.

- 1 In the editor pick the dial - say **Menu 1** - and find **Sets of this dial** under the slot list.
- 2 Press **New set...**, call it *UV* and choose **A copy of what the dial shows now** (or **An empty dial**). The key keeps opening what it opened; the new set becomes the dial's next page, and a **tab** for it opens above the dial's contents - *Sets: Modelling (on the key) | UV*.
- 3 Change the new set like any dial - directions, list, variants. Click a tab to see and edit that set. Save.
- 4 In 3ds Max hold **Alt+1** and **turn the mouse wheel**: the dial shows its next page (wheel towards you) or the previous one, in the same place. The footer says *page 2 of 2 - mouse wheel*. Release over a command to run it. The next press opens the page you turned to last.



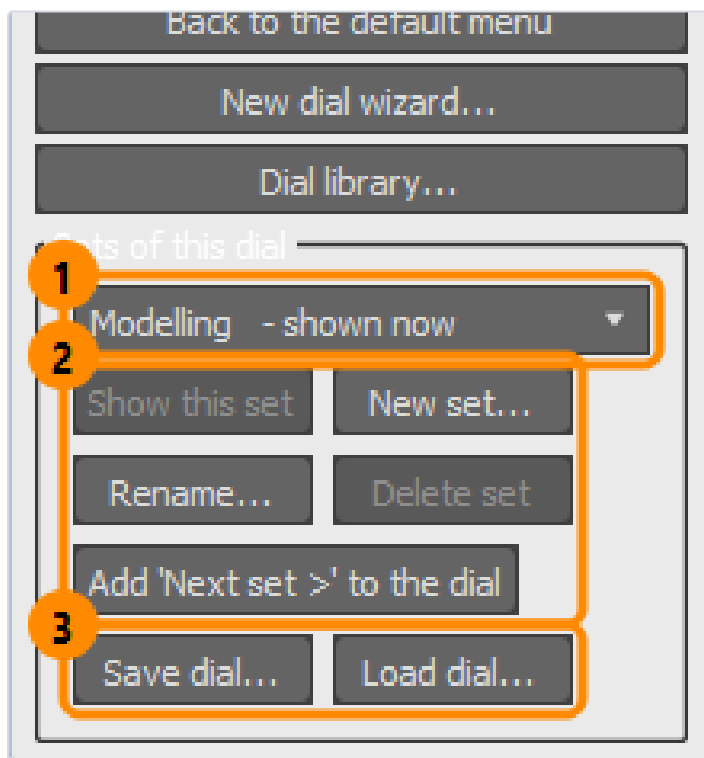
The sets of a dial as tabs (1); Put this set on the key (2).

TRY IT

Hold Alt+1, give the wheel one notch towards you, aim at a command of the UV page and release. Press Alt+1 again: it is Modelling once more.

To change what the key opens, any of these:

- 1 In the editor: open the set's tab and press **Put this set on the key** - or pick the set in **Sets of this dial** and press **Show this set**. Save.
- 2 From the dial itself: press **Add 'Next set >' to the dial**. Every set of the dial gets a *Next set >* row under the dial - pick it and the key opens the next set from then on. Save.
- 3 With a key: **Customize > Hotkey Editor**, search `next set`, give **Menu 1 - next set** a key. There is one such action for each of the 24 dials.



Sets of this dial: the sets (1), switching and managing them (2), saving and loading one dial (3).

The wheel is taken only when it can turn a page - while you hold the key of a dial with more than one set, at its top level. In a submenu, in the hotbox, in the search or on a dial with one set it zooms and scrolls as always. One notch turns one page, also on a fine wheel or a touchpad.

The switch from the dial or a key writes the file at once and the dial changes in a moment. With the editor open and unsaved changes in it, press **Reload from file** first - Save would otherwise find the file changed and offer a copy.

21

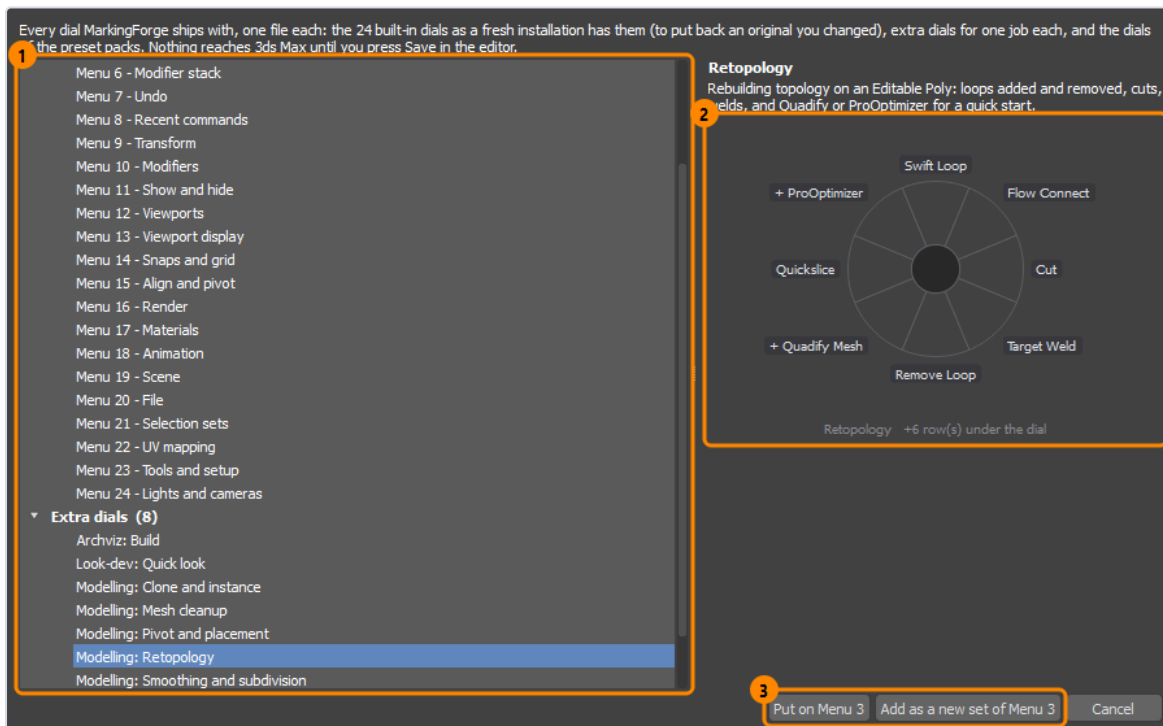
One dial from a file, or from the library

3 min · You will learn: to load a single dial, share one, and get an original back

Every dial can live in a file of its own (`.mfdial`). MarkingForge ships 47 of them - the 24 built-in dials as a fresh installation has them, eight extra dials for one job each and the 15 dials of the preset packs - in the **dial library**.

A dial from the library

- 1 Pick a dial in the slot list and press **Dial library...** under it.
- 2 Its built-in original is selected: **Restore Menu N** puts it back as it was shipped.
- 3 Or choose another dial - an extra one (Mesh cleanup, Pivot and placement, Clone and instance, Smoothing and subdivision, Retopology, Quick look, rigging Links and helpers, archviz Build) or one of the preset packs' - and press **Put on Menu N** (the key and mouse button stay), or **Add as a new set of Menu N** to keep what the dial shows: the key keeps opening it and the new dial is the next page under the mouse wheel.
- 4 Save.



The dial library: the dials by group (1), the one chosen (2), Put on Menu N and Add as a new set of Menu N (3).

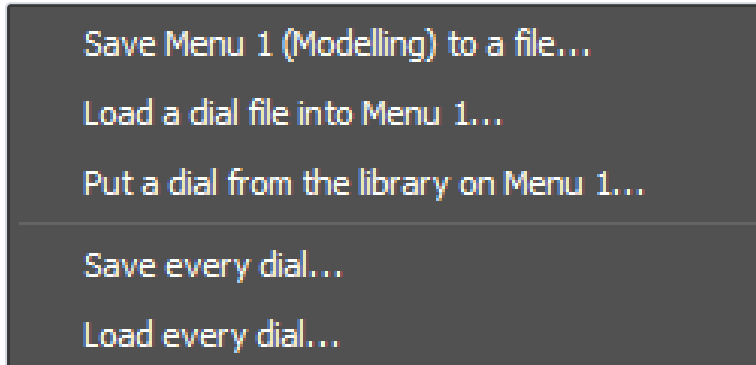
A dial from a file

- 1 Pick the dial and press **Load dial...** in **Sets of this dial**, and choose the `.mfdial` file.
- 2 If the file holds scripts, **Remove the scripts** is the default - keep them only for a file you trust.
- 3 Choose **A new set (keeps what the dial shows now)** or **Replace what the dial shows**. Save.

Save dial... beside it writes the set chosen in the list (it follows the open tab) to a `.mfdial` file, to keep it or send it.

Right-click a dial

The same is one click away where the dial is: right-click it **in the list of dials** or **on its drawing** beside the form.



Right-click on Menu 1: save it, load a file into it, put a library dial on it - or save or load every dial.

22

A new dial in a minute - the wizard and ready-made variants

4 min · You will learn: the quickest ways to a working dial of your own

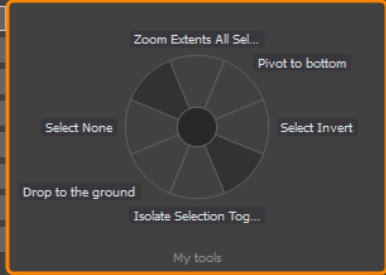
The new-dial wizard

- 1 Press **New dial wizard...** under the slot list.
- 2 Pick the dial and a name. Leave **Add it as a new set of this dial - the key keeps what it opens now** on: what the dial shows now is kept, and your new dial is its next page.
- 3 For each direction type part of a command's name and pick it from the list - 3ds Max commands and the ready-made scripts are both there. The drawing beside the fields shows the dial as you fill it; a name that matches nothing turns red.
- 4 Finish, then Save.

2. Eight commands

Type part of a command's name and pick it from the list - 3ds Max commands and the ready-made scripts. Leave a direction empty to keep it free. Opposite directions for opposite commands are the easiest to remember.

N	Zoom Extents All Selected [Main UI]
NE	Pivot to bottom [Script library]
E	Select Invert [Main UI]
SE	(empty)
S	Isolate Selection Toggle [Main UI]
SW	Drop to the ground [Script library]
W	Select None [Main UI]
NW	(empty)



< Back Next > Cancel

The wizard's second page: eight fields and the dial they make.

Ready-made variants

- 1 Pick a dial and press **From a template...** next to **Add a variant...**
- 2 On the left, every context variant the built-in dials have - Editable Poly level by level, Edit Poly, splines, cameras, lights, the windows - and each built-in dial as a whole. On the right, when it appears and what it holds.
- 3 Press **Add as a variant**. It comes with its condition; a whole dial asks for one. Change anything afterwards, then Save.

Ready-made variants from the built-in dials: a set of commands together with the situation it appears in. Pick one - a copy is added to the dial you are editing, and you can change both afterwards.

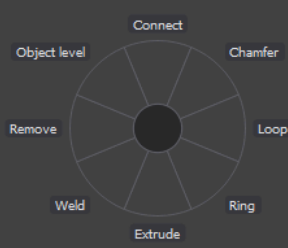
▼ Menu 1 - Modelling

- (the whole dial)
- Materials (Slate)
- Track View
- Particle View
- Scene Explorer
- Modifier stack
- UV (Unwrap)
- Edit Poly: vertex
- Edit Poly: edge
- Edit Poly: border
- Edit Poly: polygon
- Edit Poly: element
- Edit Poly: object
- Poly: vertex
- Poly: edge**
- Poly: border
- Poly: polygon
- Poly: element
- Poly: object
- Spline: vertex
- Spline: spline
- Spline
- Nothing selected

Menu 2 - Hotbox - Max's menu

► Menu 3 - Create

Appears when class = Editable Poly and subobject: 2 - usually edge



Poly: edge +8 row(s) under the dial

Add as a variant Cancel

From a template: the variants of the built-in dials, with their conditions.

Learning plan and good habits

When	Do this
Day 1	Use Alt+1, Alt+4, Alt+5 and Alt+7 with the dial visible. Print the cheat sheet.
Days 2-5	Start flicking the five gestures of tutorial 3. Add Ctrl+Alt+1 (Transform) and Ctrl+Alt+3 (Show and hide).
Week 2	Replace directions you never use with commands you reach for in menus. Add one context variant for your most common task.
Week 3	Colour your most used dials - a ready-made look is one click. Put the rest of your frequent commands in lists; promote the ones your hand wants.
Week 4	Give the dial of your main task a second page for a job you do in bursts - UV, retopology, a dial from the library - and turn to it with the mouse wheel.

Rules for your own dials

Opposites opposite

Grow / Shrink, Hide / Unhide, Open / Save across the dial - one movement and its reverse.

Keep places stable

Do not move a command once your hand knows it. Add, do not reshuffle.

One question per dial

"What do I create", "what do I hide" - a dial of unrelated commands is hard to learn.

Specific variants first

Edit Poly at vertex level above Edit Poly, above Editable Poly.

Short labels

One or two words. The hand needs the direction, the eye only a hint.

Test with a flick

If a gesture only works while looking, the direction is in the wrong place.

If something does not work

What you see	What to do
Holding the key does nothing	MarkingForge > Set up standard shortcuts... , or check the key in Customize > Hotkey Editor .
A different menu than expected	A context variant matched - the object under the cursor, the sub-object level or the window decides. Move the cursor to empty space and try again.
A red caption	That command does not exist in this 3ds Max (a missing plugin). Replace it in the editor.
An empty dial	A live dial with nothing to show - select one object for the modifier stack, create named selection sets for Selection sets.
My change is not on the dial	Press Save in the editor.
I want the original dials back	Editor > Load default dials...
I changed one dial and want its original back	Pick it in the editor, Dial library... , Restore Menu N , Save.
Turning the mouse wheel over a dial zooms the viewport	The dial has one set, or you are in a submenu, the hotbox or the search - the wheel turns pages only at the top level of a dial with more than one set (tutorial 20).
A dial stayed on screen after you let go, and other dial keys did nothing	3ds Max did not report the release (seen after starting a Text object). Since 1.4.4 MarkingForge watches the key itself: a quarter of a second later the dial closes exactly as a release would have closed it, and MarkingForge > Diagnostics > Event log shows a RELEASE_LOST line. If a dial still stays, press Esc and send that log with your report.
Letters typed over a dial vanish	That is Type-to-search (trick 5). Esc closes the search; switch it off in Experimental features if you do not use it.
After switching panels a dial key does nothing until you click a viewport	The keyboard was left on 3ds Max's menu bar (a lone Alt does that). Since 1.0.9 MarkingForge hands it back by itself. If it still happens, press Esc or click the viewport, then run MarkingForge > Diagnostics > Event log and send the lines marked FOCUS with your report.
Where are the other guides?	MarkingForge > Documentation opens the folder with all of them.

Everything else - every option, every command, the configuration file and more problems with their fixes - is in the **MarkingForge Reference**.

Still stuck? Open an issue at github.com/looki666/MarkingForge/issues or write to forgeplugins@gmail.com.
MarkingForge > Report a bug... puts your 3ds Max and MarkingForge details on the clipboard, ready to paste.

Support, bug reports and feature requests

ISSUES	github.com/looki666/MarkingForge/issues
E-MAIL	forgeplugins@gmail.com
WEB	https://forgeplugins.tech (forgeplugins.tech - more plugins by the author)

Bug reports and feature requests are best opened as issues on GitHub - the forms there ask for exactly what is needed, and you can follow the fix. Questions, and anything you would rather not post publicly (licence, purchase), go to the e-mail address. Every message is read.

Reporting a bug

- 1 In 3ds Max choose **MarkingForge > Report a bug...** - it copies your 3ds Max version, MarkingForge version, configuration summary and start-up log to the clipboard. It works even when the plugin did not load.
- 2 Open github.com/looki666/MarkingForge/issues and pick **Bug report** - or answer **Yes** in the dialog to write an e-mail to forgeplugins@gmail.com instead.
- 3 Say what you did, what you expected and what happened instead, then paste the details with **Ctrl+V**.
- 4 Add a screenshot if you can. When the problem is in one dial, attach a preset of it (editor > **Save preset...**).

Suggesting a feature

Open an issue with the **Feature request** form, choose **MarkingForge > Suggest a feature...**, or simply write. Describe what you would like to do and how you do it today - a concrete example from your work says more than a list of wishes. Ideas for new built-in dials, commands and options are all welcome.

Nothing is sent by the plugin itself: the details go to the clipboard and your own e-mail program, so you see everything that leaves your computer. MarkingForge has no network access.